

The Measurement Protocol Decides the Conclusion: A Methodological Study of CNN and Vision Transformer Adversarial Robustness Comparisons

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Abstract—Comparisons of adversarial robustness across architecture families report conflicting conclusions, and the disagreement is usually attributed to models, training recipes, or datasets. We show that a substantial part of it comes from the measurement protocol itself. Using adversarially trained ResNet-18 and ViT-Tiny pairs, three runs per architecture, full-test-set evaluation with per-sample logging, and a validation split held out before any training, we hold models, data, threat model and attack budget fixed and vary only how robustness and transfer are scored. Across four established conditioning protocols the measured CNN-to-ViT transfer asymmetry ranges from 4.4 to 19.4 points on CIFAR-10, whereas retraining both architectures from scratch moves the same quantity by under 1.5 points, and the protocol decides whether an equivalence test is accepted or rejected. Replicating the design on CIFAR-100 leaves the spread in the same range and preserves the sign in all twelve measurements. On SVHN, where the same recipe leaves the architectures only 1.85 points apart in clean accuracy, two protocols report a CNN advantage and two a ViT advantage: once the models are matched, the measurement choice decides not only the size of the asymmetry but its sign. A three-architecture control shows that the deviation of unconditioned from conditioned rates is algebraically equal to the target’s clean error, not merely correlated with it, so raw rates are contaminated by target weakness by construction. Holding the training trajectory fixed and varying only the checkpoint-selection rule adds a fourth, almost never reported, source of variance. Three claims common in this area also fail to replicate. We close with reporting requirements for cross-architecture robustness studies.

Index Terms—Adversarial robustness, convolutional neural networks, evaluation methodology, deep learning security, reproducibility, transfer attacks, Vision Transformer

I. INTRODUCTION

Deep neural networks are vulnerable to adversarial examples, carefully crafted perturbations that cause confident misclassification while remaining bounded by a small norm budget [1], [2]. As Vision Transformers (ViTs) [3], [4] joined convolutional neural networks (CNNs) [5] as a dominant architecture family, a natural question followed: which family is more robust, and why?

The literature answers this question inconsistently. Bai et al. [6] report that under matched training recipes transformers do not surpass CNNs, whereas Benz et al. [7] report ViTs to be more robust for standardly trained models; studies of

cross-architecture transfer likewise disagree on whether attacks move more easily from CNNs to ViTs or the reverse [8]–[10]. These disagreements are usually attributed to differences in models, training recipes, or datasets, and the standard response is to compare under a matched recipe.

This paper makes a different observation. Even when the models, the data, the threat model, and the attack budget are all held fixed, the conclusion still depends on a choice that is rarely reported: the *measurement protocol*. Transfer studies must decide which samples enter the denominator (all test samples, those the target classifies correctly, those both models classify correctly, or those on which the source attack succeeded), and robustness comparisons must decide whether to report an absolute score or to separate it into clean accuracy and conditional susceptibility. These choices are usually treated as reporting details. We show that they are part of the result.

Our setting is deliberately narrow so that the measurement question is isolated. We adversarially train ResNet-18 and ViT-Tiny on CIFAR-10 [11] under a matched objective and threat model, with three independent training runs per architecture and a fixed validation split held out before *any* training stage, so that no model is selected on data it has seen. All evaluations run on the full test set with per-sample outcome logging, which makes paired statistics possible: the same samples can be re-weighted under different protocols without re-running an attack. We then vary the protocol, not the models, and measure how far the conclusion moves.

We also treat three claims that recur in architecture-comparison papers as measurable hypotheses rather than as interpretation: that CNN input gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that ViT-specific transfer attacks improve cross-architecture transfer. Each is tested directly with a purpose-built measurement.

Our contributions are as follows. We first quantify protocol sensitivity. On identical models and data, the measured cross-architecture transfer asymmetry ranges from +4.4 to +19.4 points across four established conditioning protocols, a 4.4-fold spread, whereas retraining both architectures from scratch moves the same quantity by under 1.5 points. The protocol decides whether an equivalence test is accepted or rejected, while the direction of the asymmetry, unlike its magnitude, is stable. Repeating the full design on CIFAR-100 leaves the mean per-seed protocol spread at 13.83 ± 1.30 points, close to the CIFAR-10 value, and preserves the sign in all twelve mea-

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surements. Holding a training trajectory fixed and varying only the offline checkpoint-selection rule moves reported PGD-10 accuracy by 1.58 to 2.85 points, which places selection alongside seeds, attacks, and protocols as a reportable source of variance.

We then show that unconditioned transfer rates are confounded, and measure by how much. In a 3×3 transfer matrix that adds a WideResNet-28-10 reference, the deviation of raw from conditioned rates is not an empirical correlation but an identity. Because samples the target already misclassifies are essentially never rescued by the perturbation (0.989 to 1.000 over 36 directions), the raw rate equals the clean error plus the conditioned rate scaled by one minus that error, to within 0.41 points. Incoming transfer also tracks the target’s own vulnerability rather than the source architecture ($r = 0.986$ across three targets), which reframes the asymmetry as a property of the weaker target rather than of stronger CNN-crafted attacks. A conditional decomposition completes the picture: absolute robustness factors exactly into clean accuracy and conditional susceptibility, and reporting both matters, because the attribution between them changed between our earlier single-run checkpoints and three fresh runs while the absolute ordering did not.

We report three negative results. Scale-invariant sparsity differences do not correspond to spatial localization differences, on four spatial measures of which three are null; CLS attention entropy does not change measurably under attack at $n = 1000$; and Token Gradient Regularization [12], a ViT-specific transfer attack, gives no transfer gain in this regime, losing to plain MI-FGSM under a matched budget. Two behavioral differences do survive. Gradient sparsity differences reproduce across independent checkpoint sets, are created by adversarial training rather than by architecture, and are accompanied by an alignment difference that exists only in absolute cosine. Adversarial damage accumulates at qualitatively different depths in the two architectures, and the measured drift profile depends on the token aggregation used to compute it.

We close with a short set of reporting requirements that follow from these measurements.

A preliminary version of this study was submitted to ATEEC 2026 [13]. That version reports the conditional decomposition, the four-protocol transfer comparison, the gradient structure measurements and the block-wise drift profiles on CIFAR-10 alone, and describes them as preliminary. The present paper adds two further datasets, a second threat model, the algebraic identity of Section IV, the checkpoint-selection ablation, the transformer-specific transfer attack, and the pre-registered analysis protocol, and it corrects one claim of the earlier version: the direction of the asymmetry is not stable once the architectures are matched in clean accuracy.

Section II reviews adversarial evaluation practice and architecture-comparison studies. Section III describes the training protocol, the conditioning protocols, and the measurements. Section IV reports results. Section V discusses implications and limitations, and Section VI concludes.

II. RELATED WORK

A. Adversarial Attacks

Adversarial attacks aim to find minimal perturbations that cause misclassification. The Fast Gradient Sign Method (FGSM) [2] generates perturbations in a single step by following the sign of the loss gradient, with the perturbation magnitude bounded under the L_∞ norm. Projected Gradient Descent (PGD) [14] extends FGSM through iterative optimization, taking multiple gradient-sign steps and projecting back onto the ϵ -ball after each step (formal definitions in Section III-C).

The Carlini & Wagner (C&W) attack [15] formulates adversarial example generation as an optimization problem, often achieving higher success rates but at increased computational cost. More recently, AutoAttack [16] combines multiple attack strategies (APGD-CE, APGD-T, FAB-T, Square Attack) to provide a reliable robustness evaluation, becoming the de facto standard for benchmarking.

B. Defense Mechanisms

Adversarial training (AT) [14] remains the most effective defense, training models on adversarial examples generated during each batch:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\|\delta\|_\infty \leq \epsilon} \mathcal{L}(f_\theta(x + \delta), y) \right] \quad (1)$$

TRADES [17] provides a theoretically-grounded trade-off between natural accuracy and robustness:

$$\mathcal{L}_{\text{TRADES}} = \mathcal{L}_{CE}(f(x), y) + \beta \cdot \text{KL}(f(x) \| f(x_{adv})) \quad (2)$$

where β controls the trade-off parameter.

MART [18] addresses the issue of misclassified examples by emphasizing them during training. Other approaches include input transformation [19], certified defenses via randomized smoothing [20], and ensemble methods [21]. Recent advances in certified robustness include Adaptive Randomized Smoothing [22], which improves certification radii through multi-step defenses.

C. CNN Robustness

Extensive research has characterized the adversarial robustness of CNN architectures. Madry et al. [14] demonstrated that adversarial training with PGD attacks provides strong empirical robustness. The RobustBench leaderboard [23] tracks state-of-the-art CNN robustness, with WideResNet variants achieving the highest robust accuracy on CIFAR-10.

Studies have shown that model capacity correlates with achievable robustness [14], with wider and deeper networks demonstrating improved robust accuracy. Architecture-specific findings include the role of skip connections in adversarial transferability [24] and the distinct batch-normalization statistics that adversarial training induces at scale [25]; relatedly, auxiliary batch-normalization branches allow adversarial examples to improve clean accuracy [26]. Adversarial training is also known to reshape input-gradient structure itself, yielding more perceptually aligned [27] and sparser [28] gradients.

D. Vision Transformer Robustness

Vision Transformers [3], building on the self-attention mechanism introduced by Vaswani et al. [4], process images as sequences of patches; their internal representations differ measurably from CNNs’ layer by layer [29]. Initial studies suggested that ViTs exhibit some inherent robustness properties due to their global receptive field [30]. Bai et al. [6] systematically investigated whether transformers are more robust than CNNs, finding that under fair training recipes transformers are not more adversarially robust than CNNs, while exhibiting superior robustness under distribution shift attributed to self-attention; Benz et al. [7] reported an early direct comparison including MLP-Mixer and linked the differences to frequency preferences. Our work extends this line of inquiry by analyzing adversarially trained models of comparable scale and providing behavioral characterizations through gradient characteristics and feature degradation analysis. Subsequent work revealed that ViTs remain vulnerable to adversarial attacks, particularly gradient-based methods [8], with patch-targeted attacks exploiting the attention mechanism directly [31] and attention shown to shift measurably under patch perturbations [32].

Shao et al. [33] demonstrated that adversarial training can improve ViT robustness and that ViT features rely less on high-frequency content, though challenges remain due to training instability. ViT robustness to common corruptions and occlusions has also been characterized [34]. More recently, Jain et al. [35] proposed Adaptive Attention Scaling Adversarial Training (AAS-AT) to address ViT-specific training challenges, achieving state-of-the-art robustness on CIFAR-10 and ImageNet-100; with diffusion-augmented adversarial training, ViTs have even been reported to surpass WideRes-Nets on small-scale datasets [36]. Zhu et al. [37] strengthened transferable attacks on ViTs through gradient normalization scaling and high-frequency adaptation. Despite these advances, comprehensive behavioral comparisons between CNN and ViT robustness under unified evaluation frameworks remain limited.

E. Transfer Attacks

Adversarial examples often transfer between models, enabling black-box attacks [38]. The transferability depends on factors including model architecture, training data, and attack methods [39]. Foundational transfer studies already evaluate transferability on correctly classified samples [40], and standard attack papers condition their success rates for the same reason [41]; recent evaluation guidelines make such protocol choices explicit [42]. Two recent benchmarks move in the same direction: TA-Bench [43] standardizes the comparison of more than thirty transfer attacks over a common pool of substitute and victim models, and Yu et al. [44] report that transferability is systematically overestimated when it is measured across a single architecture family and propose three evaluation protocols in response. Both hold the scoring rule fixed and vary the attack and the model pool. We do the opposite: the models, the data and the attack budget are fixed, and only the scoring rule changes, which isolates the

contribution of the protocol itself and makes it possible to state the arithmetic relation between the resulting estimates. In ensemble design, Ravikumar et al. [9] restrict transfer measurement to samples correctly classified by both models precisely to account for accuracy differences between them, the same confound our conditioned metric controls.

Cross-architecture transfer between CNNs and ViTs has received increasing attention. Mahmood et al. [8] measured cross-architecture transfer with a both-correct protocol and found notably low CNN $\leftrightarrow$ ViT transferability; Naseer et al. [10] showed that adversarial examples crafted *on* ViTs transfer poorly under conventional attacks and proposed self-ensemble and token refinement to improve ViT-sourced transferability, and Wei et al. [45] improved ViT-sourced transfer by bypassing attention gradients and patch-wise dropout. Building on these, Token Gradient Regularization (TGR) [12] crafts transferable adversarial samples by exploiting ViT’s attention components, while Momentum Integrated Gradients (MIG) [46] achieves improved transferability across both ViTs and CNNs. Chen et al. [47] systematically analyze the factors affecting cross-architecture transferability, finding that architectural inductive biases, with CNNs preferring high-frequency information while ViTs favor low-frequency patterns, significantly impact transfer success. Waseda et al. [48] separate transfer outcomes further, by whether the target reproduces the source’s exact error or fails in a different way, and report that both occur even between closely related models. Understanding these transfer patterns is crucial for developing robust ensemble systems and evaluating real-world attack scenarios.

F. Research Gap

While individual studies have examined CNN and ViT robustness separately, and the pitfalls of unfair CNN and transformer comparisons have been highlighted before [6], [49], existing comparative work typically reports robustness scores in isolation, often with inconsistent perturbation budgets, attack configurations, or model scales, rather than pairing the comparison with behavioral analysis. A recent study in this journal compares ViT and CNN adversarial robustness directly [50]; our work differs by evaluating on the full test set with AutoAttack and paired statistics, by controlling transfer measurement for clean-accuracy differences, and by accompanying the comparison with gradient- and feature-level behavioral analyses.

The evaluation-methodology literature has repeatedly shown that how robustness is measured decides what is concluded: Carlini et al. [51] catalogue the failure modes of robustness evaluation, and Tramèr et al. [52] show that defenses judged robust under fixed attacks fall to adaptive ones. That work concerns attack strength, that is whether the measurement is strong enough. The question we ask is orthogonal and, as far as we can tell, unmeasured: with attack strength held fixed, how much does the *scoring rule* alone move the conclusion? Our work engages that question directly and makes it the object of study. Conditioning transfer evaluation on correctly classified samples is established practice [8], [9], [40], [41], and individual papers pick one protocol and proceed. What

is missing is a controlled estimate of how much that pick matters. We supply it by holding the models, the data, the threat model, and the attack budget fixed and varying only the protocol, and we find that the measured asymmetry moves by 15.01 ± 0.84 points across the four protocols, a 4.4-fold spread, while retraining both architectures moves that same quantity by under 1.5 points. We further show that the confound in unconditioned rates is not a tendency but an identity ($r_{\text{raw}} - r_{\text{cond}} = e(1 - r_{\text{cond}})$, verified to within 0.41 points over 36 directions), which turns a known qualitative caveat into a correction that can be applied directly to published results.

Three further claims recur in this literature that our design lets us test rather than assume: that CNN gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that transformer-specific transfer attacks [10], [12] improve cross-architecture transfer. We measure each directly, and none holds in our setting, and we report these results because an untested assumption of this kind is exactly what a protocol-sensitive comparison cannot afford.

III. METHODOLOGY

A. Threat Model

We consider the standard L_∞ threat model where an adversary can perturb input images within a bounded region:

$$\mathcal{B}_\epsilon(x) = \{x' : \|x' - x\|_\infty \leq \epsilon\} \quad (3)$$

Following established conventions for CIFAR-10 [11], [23], we use $\epsilon = 8/255 \approx 0.031$ as the primary perturbation budget, with additional full-test PGD-10 evaluations at $\epsilon \in \{2/255, 4/255, 16/255\}$ for the epsilon sweep analysis.

We evaluate under both white-box (full model access) and black-box (transfer attack) scenarios. For white-box attacks, the adversary has complete knowledge of model architecture and parameters. For transfer attacks, adversarial examples are generated on a source model and evaluated on a different target model.

B. Model Architectures

a) *CNN Models: ResNet-18* [5]: A widely-used baseline CNN with residual connections, containing 11.2M parameters. We use the standard architecture designed for CIFAR-10 (initial 3×3 convolution instead of 7×7).

WideResNet-28-10 [53]: A wider variant of ResNet with 36.5M parameters, representing the state-of-the-art architecture for adversarial robustness on CIFAR-10. We use the pretrained robust checkpoint of Goyal et al. [54] distributed via RobustBench [23]; note that this model was trained with additional unlabeled data and a stronger recipe than our standard adversarial training, so it serves as a reference point rather than a matched comparison.

b) *Vision Transformer Models: ViT-Tiny*: We use the ViT-Tiny architecture from the `timm` library [55], adapted for CIFAR-10 by bilinearly upsampling inputs to 224×224 . The architecture uses 16×16 patches (196 patches per image), 192-dimensional embeddings, 12 transformer blocks with 3 attention heads each, MLP ratio of 4, and totals 5.7M parameters. The model is trained *from scratch* on CIFAR-10 (no

ImageNet pretraining) with basic augmentation (random crop and horizontal flip), then adversarially fine-tuned. Since ViT robustness on small datasets is known to be sensitive to the training recipe [56], [57], our architecture-level conclusions should be interpreted as holding *under this baseline training recipe* rather than for optimally trained ViTs.

C. Attack Methods

a) *FGSM*: Single-step attack with step size equal to the perturbation budget:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}_{CE}(f(x), y)) \quad (4)$$

b) *PGD*: Iterative attack with step size $\alpha = 2/255$ and random initialization:

$$x^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)}(x^t + \alpha \cdot \text{sign}(\nabla_{x^t} \mathcal{L}_{CE}(f(x^t), y))) \quad (5)$$

with $x^0 = x + \mathcal{U}(-\epsilon, \epsilon)$. We use PGD-10 (10 iterations) for both adversarial training and evaluation throughout this work (Table I).

c) *AutoAttack*: We use AutoAttack [16], the standardized ensemble attack combining APGD-CE (Auto-PGD with cross-entropy loss), APGD-T (Auto-PGD with targeted DLR loss), FAB-T (targeted Fast Adaptive Boundary attack), and Square Attack (score-based black-box attack). This ensemble is a standardized strong benchmark for robustness evaluation.

D. Defense Methods

a) *Adversarial Training (AT)*: We employ a mixed adversarial training objective that combines clean and adversarial losses:

$$\mathcal{L} = (1 - \lambda) \mathcal{L}_{CE}(f_\theta(x), y) + \lambda \mathcal{L}_{CE}(f_\theta(x_{adv}), y) \quad (6)$$

where $\lambda = 0.5$ balances standard classification performance with adversarial robustness. The inner maximization generates adversarial examples using PGD-10 ($\alpha = 2/255$, $\epsilon = 8/255$), computed with the model in training mode so that attack generation and the parameter update see consistent batch-normalization statistics, following standard adversarial-training implementations [58]. This mixed objective, widely adopted in practice, helps maintain clean accuracy while building robustness, as training solely on adversarial examples can degrade natural performance. While alternative formulations such as TRADES [17] offer theoretically grounded trade-off mechanisms, this work focuses on standard adversarial training to isolate architectural effects from defense-specific interactions.

E. Training Protocol

a) *Clean Training*: For CNNs, we use SGD optimizer with initial learning rate 0.1, cosine annealing schedule [59], weight decay 5×10^{-4} , and 200 epochs. For ViTs, we use AdamW optimizer [60] with initial learning rate 10^{-3} , cosine annealing, weight decay 0.05, and 200 epochs.

b) *Adversarial Training*: Starting from our clean-trained checkpoints, we fine-tune with reduced learning rates: CNNs use SGD with LR 10^{-3} and cosine annealing for up to 100 epochs, while ViTs use AdamW with the same learning rate and schedule. Gradients are clipped to a global L_2 norm of 10, and batches with non-finite loss are skipped. Data augmentation includes random cropping with 4-pixel padding and horizontal flipping; all models consume raw pixel inputs in $[0, 1]$ without mean/std normalization, and attacks and epsilon budgets operate directly in this pixel space. The training batch size is 128 for the CNN and 64 for the ViT (evaluation batch sizes vary by analysis). Early stopping uses patience 20 with a 0.1 percentage-point improvement threshold.

c) *Validation Split and Repetitions*: Model selection and early stopping use PGD-10 adversarial accuracy on a fixed 2,000-sample validation split. The split is drawn once (seed 777, indices stored with the code) and is shared by every model in the study, and it is excluded from the training set of *both* the clean and the adversarial stage, so no checkpoint is ever selected on data it was trained on. The test set is touched only by the final post-hoc evaluations. We repeat the full pipeline, clean pre-training followed by adversarial fine-tuning, three times per architecture with independent seeds (1001 to 1003 for ResNet-18, 2001 to 2003 for ViT-Tiny), and pair the runs by index so that every cross-architecture comparison is between models trained under the same repetition. All tables report the mean and standard deviation over these three runs.

For one comparison in Section IV-A we also refer to an earlier pair of checkpoints trained before this protocol was adopted, whose validation split had been part of clean pre-training. We report them explicitly as a contrast, not as results, because the difference between the two checkpoint sets is itself evidence about how far a single-run conclusion can move.

F. Evaluation Metrics

a) *Robustness Metrics*: We evaluate models using three standard metrics: clean accuracy (classification performance on unperturbed test images), robust accuracy (classification performance on adversarial examples), and attack success rate (fraction of correctly classified clean images that become misclassified after attack). For white-box evaluations we additionally report the *conditioned fooling rate*, the fraction of clean-correct samples that the attack flips, and its complement, conditioned survival. Because both PGD and AutoAttack count clean-misclassified samples as non-robust in our per-sample logs, absolute robust accuracy factorizes exactly as $\text{robust} = \text{clean} \times \text{conditioned survival}$, which lets us decompose robustness differences into a clean-accuracy component and a conditional-susceptibility component. Finally, we report accuracies on the *both-correct* subset (samples that both compared models classify correctly on clean inputs), which yields an exactly paired comparison on a shared sample set.

b) *Transfer Attack Metrics*: Our primary transfer metric is the *conditioned fooling rate*: among the samples that the target model classifies correctly on clean inputs, the fraction that becomes misclassified under adversarial examples crafted on the source model. Given adversarial examples $\{x_{adv}^S\}$

generated against source model f_S , the conditioned transfer rate to target model f_T is:

$$T_c(f_S \rightarrow f_T) = \frac{\sum_{i=1}^n \mathbf{1}[f_T(x_i) = y_i] \mathbf{1}[f_T(x_{adv,i}^S) \neq y_i]}{\sum_{i=1}^n \mathbf{1}[f_T(x_i) = y_i]} \quad (7)$$

Conditioning transfer evaluation on correctly classified samples follows established practice [8], [9], [40]–[42] and is essential in cross-architecture comparisons: the raw misclassification rate $\frac{1}{n} \sum_i \mathbf{1}[f_T(x_{adv,i}^S) \neq y_i]$ includes the target’s clean error, so a target with lower clean accuracy appears more “vulnerable” to transfer even when no additional samples are flipped by the attack. We report the raw rate as a secondary metric for transparency and to quantify the size of this confound. Because the conditioning set itself is a protocol choice, we additionally report two established alternatives: the *both-correct* protocol, which conditions on samples that both the source and the target classify correctly [8], [9], and the *successful-source* protocol, which conditions on target-correct samples whose attack succeeded on the source in the white-box sense. As Section IV-B shows, these protocols can yield qualitatively different asymmetry conclusions, so we treat the protocol as part of the reported result rather than an implementation detail. Transfer rates are computed on the full test set (10,000 samples) with per-sample outcomes logged, and we report 95% confidence intervals obtained by percentile bootstrap over samples (10,000 resamples).

c) *Gradient Analysis Metrics*: We analyze gradient properties through three complementary metric families, computing gradients of per-sample losses so that magnitudes are independent of the evaluation batch size. Gradient norm ($\|\nabla_x \mathcal{L}\|_2$ and $\|\nabla_x \mathcal{L}\|_\infty$) measures overall sensitivity magnitude. Gradient sparsity is quantified with scale-invariant measures: Hoyer sparsity, the Gini coefficient of $|g|$, and the fraction of components below 1% of the per-sample maximum, since fixed absolute thresholds conflate gradient scale with structure; Hoyer and Gini satisfy the scale-invariance axioms formalized by Hurley and Rickard [61]. Gradient alignment measures the average pairwise absolute cosine similarity across samples (computed over all pairs of the 500 analysis samples):

$$\text{Alignment} = \frac{1}{N(N-1)} \sum_{i \neq j} \left| \frac{\nabla_{x_i} \mathcal{L} \cdot \nabla_{x_j} \mathcal{L}}{\|\nabla_{x_i} \mathcal{L}\| \|\nabla_{x_j} \mathcal{L}\|} \right| \quad (8)$$

We use the absolute cosine because shared sensitivity *axes* matter for attack geometry regardless of sign; since sign coherence is what universal-perturbation arguments additionally require, we also report the signed mean cosine, which turns out to be near zero for every model (Section IV-F). High absolute alignment indicates similar perturbation axes across inputs.

d) *Spatial Locality Metrics*: Sparsity measures how many gradient components are large but says nothing about where they lie in the image, so we measure spatial structure separately on the channel-summed energy map $e = \sum_c (\nabla_x \mathcal{L})_c^2$, normalized to sum to one. We report the smallest pixel fraction whose energy sums to 50% and to 90% (smaller means more localized), the Shannon entropy of e viewed as a distribution over pixels (lower means more localized), and Moran’s I with 4-neighbourhood weights, which is positive

when energy clusters in adjacent pixels. All four are invariant to the scale of the gradient, and all are computed per sample so that the two architectures can be compared with paired tests on identical inputs.

e) Token Aggregation for Layer-wise Analysis: A transformer block emits one vector per token, so summarizing a block requires a choice. We report three: the CLS token alone, the mean over the 196 patch tokens, and the concatenation of all tokens flattened per sample. For the CNN each residual block output is flattened over its spatial dimensions. We report all three for the ViT because, as Section IV-G shows, they give different drift magnitudes and place the minimum at different depths.

f) Attention Metrics: For the ViT we extract the post-softmax CLS-to-patch attention row of each block, averaged over heads. We report its Shannon entropy, which measures how broadly attention is spread, and its displacement under attack, measured as the total variation distance between the clean and adversarial attention distributions (bounded by 1). Both are computed per sample and averaged, on $n = 1000$ samples per run.

g) ViT-Specific Transfer Attack: To test whether the transfer results depend on using architecture-agnostic attacks, we additionally generate source examples with Token Gradient Regularization (TGR) [12], which zeroes the extreme-valued entries of the token-gradient tensors in the attention, QKV, and MLP paths during backpropagation and scales the remaining gradient by $\gamma = 0.5$. Our implementation adapts the method to our CIFAR-10 ViT wrapper and to the budget used throughout this paper ($\epsilon = 8/255$, $\alpha = 2/255$, 10 steps, momentum 1.0) rather than the original ImageNet defaults; we therefore report it as an adaptation. Plain MI-FGSM [41] examples are generated in the same run on the same samples, giving a paired control at an identical budget.

h) Feature Degradation Metrics: To analyze how adversarial perturbations affect ViT representations across layers, we compute three metrics at each layer ℓ . Feature cosine similarity measures directional preservation of the representation:

$$\text{Sim}_\ell = \frac{h_\ell(x) \cdot h_\ell(x_{adv})}{\|h_\ell(x)\| \|h_\ell(x_{adv})\|} \quad (9)$$

where h_ℓ denotes the representation at layer ℓ . Feature norm change captures magnitude effects:

$$\Delta_\ell = \frac{\|h_\ell(x_{adv})\| - \|h_\ell(x)\|}{\|h_\ell(x)\|} \times 100\% \quad (10)$$

Feature L_2 distance ($\|h_\ell(x) - h_\ell(x_{adv})\|_2$) quantifies absolute deviation.

G. Statistical Validation

We separate four sources of variance and report each where it applies. *Training-run variance* is the one that bears on architecture comparisons: every table reports the mean and standard deviation over three independent runs per architecture (Section III-E). *Attack-initialization variance* is measured by re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed subset; the evaluated samples and their order are fixed, so the seeds vary only the random start of the attack.

TABLE I: Main robustness results on CIFAR-10 ($\epsilon = 8/255$), mean $\pm$ std over three independent training runs per architecture. AA denotes AutoAttack with the standard suite.

Model	Defense	Params	Clean	FGSM	PGD-10	AA
<i>CNN Models</i>						
ResNet-18	None	11.2M	95.25 $\pm$ 0.15	40.29 $\pm$ 1.77	0.03 $\pm$ 0.03	–
ResNet-18	AT	11.2M	85.78 $\pm$ 0.36	53.46 $\pm$ 0.08	44.11 $\pm$ 0.50	37.93$\pm$0.14
WRN-28-10	AT [†]	36.5M	89.48	70.91	66.92	62.76
<i>Vision Transformer Models</i>						
ViT-Tiny	None	5.7M	80.09 $\pm$ 0.60	6.10 $\pm$ 1.47	0.05 $\pm$ 0.05	–
ViT-Tiny	AT	5.7M	73.53 $\pm$ 0.55	36.31 $\pm$ 0.42	32.69 $\pm$ 0.22	29.14$\pm$0.40

[†]Robust checkpoint of Gowal et al. [54] via RobustBench [23]; a single external checkpoint, so no standard deviation applies. Clean/FGSM/PGD-10: our local full-test evaluations; AA: the RobustBench-reported figure (we did not rerun AutoAttack on it).

Protocol variance is the spread across conditioning protocols, which we treat as a result rather than as noise (Section IV-B). *Checkpoint-selection variance* is the spread produced by the offline rule that picks the final checkpoint from a fixed trajectory, namely validation split, early-stopping patience, and curve smoothing, measured over eighteen combinations while the trajectory itself is held constant. Section IV-I compares their magnitudes.

Paired tests are used wherever the same samples are scored under two conditions: McNemar’s exact test for paired binary outcomes, Wilcoxon signed-rank with Holm correction for the three sparsity measures, paired bootstrap confidence intervals (10,000 resamples) for differences, and a sign-flip permutation test (20,000 permutations) for the both-correct transfer difference. Equivalence claims use two one-sided tests (TOST). The primary margin is ± 2 points, fixed before the comparisons on the principle that differences smaller than the run-to-run variation in PGD accuracy should count as practically equivalent, and every equivalence claim is also reported at ± 1 and ± 3 points so that the verdict can be read against the margin. All evaluation and analysis scripts fix random seeds (seed 42, including CUDA determinism flags), so every reported number is reproducible from the released pipeline.

IV. EXPERIMENTS

A. Robustness Comparison

Table I presents the main robustness comparison across all evaluated models. Every cell for our models is the mean over three independent training runs per architecture, with the standard deviation over those runs; each run is evaluated on the full 10,000-sample test set.

The CNN holds an *absolute* robustness advantage under every attack: ResNet-18 AT reaches $37.93 \pm 0.14\%$ robust accuracy under AutoAttack against $29.14 \pm 0.40\%$ for ViT-Tiny AT. Because both models of a pair are evaluated on the same 10,000 samples with per-sample outcomes logged, McNemar’s paired test applies within each run; it favors the CNN decisively in all three pairs ($p < 10^{-84}$ under AutoAttack, $p < 10^{-120}$ under PGD-10). Seed-level variation is small throughout for the adversarially trained pair, at most 0.55 points on any reported quantity, so the ordering is not

TABLE II: Conditional decomposition of white-box robustness on the full test set, mean $\pm$ std over three runs. The lower block repeats the same quantities on our earlier single-run checkpoints.

Model	Cond. fooling (%)		Both-correct robust (%)	
	PGD-10	AA	PGD-10	AA
ResNet-18 AT	48.58 $\pm$ 0.80	55.78 $\pm$ 0.23	59.86 $\pm$ 0.35	51.89 $\pm$ 0.49
ViT-Tiny AT	55.53 $\pm$ 0.50	60.37 $\pm$ 0.33	46.11 $\pm$ 0.59	41.15 $\pm$ 0.39
<i>Earlier single run, validation split seen during clean pre-training (n=7,260)</i>				
ResNet-18 AT	52.15	58.16	54.92	48.15
ViT-Tiny AT	52.33	56.46	49.61	45.33

a product of run-to-run noise; the undefended baselines vary more under FGSM (up to 1.8 points), but no claim rests on them. The CNN also retains higher clean accuracy (85.78 vs 73.53%, a 12.3-point gap), reflecting the difficulty of training ViTs on small datasets from scratch; Table II separates that contribution from conditional susceptibility. AutoAttack robustness is consistently below PGD-10 robustness for both architectures (6.2 points for ResNet-18, 3.6 for ViT-Tiny), confirming the value of the stronger ensemble. The WideResNet-28-10 reference [54], at 62.76%, shows how far capacity, additional training data, and a stronger recipe jointly push robustness beyond the architectural choice; we keep it visually and analytically separate from the matched pair throughout.

Table II decomposes the absolute scores into their clean-accuracy and conditional-susceptibility components (Section III-F). Because misclassified clean inputs count as non-robust, the factorization is exact: for AutoAttack it reads $85.78 \times 0.442 = 37.93$ for the CNN and $73.53 \times 0.396 = 29.14$ for the ViT. Both factors favor the CNN. It starts from a 12.3-point higher clean accuracy *and* loses a smaller fraction of what it solves, with conditional fooling of 48.58 versus 55.53% under PGD-10 and 55.78 versus 60.37% under AutoAttack. On the jointly solved subset, the $n = 7,061 \pm 68$ samples that both models of a pair classify correctly on clean inputs and therefore an exactly paired comparison on shared inputs, the CNN leads by 13.8 and 10.7 points (McNemar $p < 10^{-84}$ in every run).

The decomposition is worth reporting even when the two factors agree, because they need not, and because their relative weight is less stable than the ordering they produce. Our first checkpoints, selected with a validation split that clean pre-training had already seen, gave the same absolute ordering but the opposite conditional one: fooling rates that were indistinguishable under PGD-10 (52.15 versus 52.33%) and slightly ViT-favorable under AutoAttack (58.16 versus 56.46%), which supported the reading that the CNN advantage was entirely a clean-accuracy effect (lower block of Table II). None of the three corrected runs reproduces that reading, and the three agree with one another far more closely than any of them agrees with the earlier run. We cannot isolate a single cause, since the corrected protocol changed the validation handling and the training run at the same time; notably, a simple data-quantity explanation does not hold, because the corrected clean models are *better* than the earlier ones despite seeing 2,000 fewer images (80.09 ± 0.60 versus 77.50% for the ViT). The

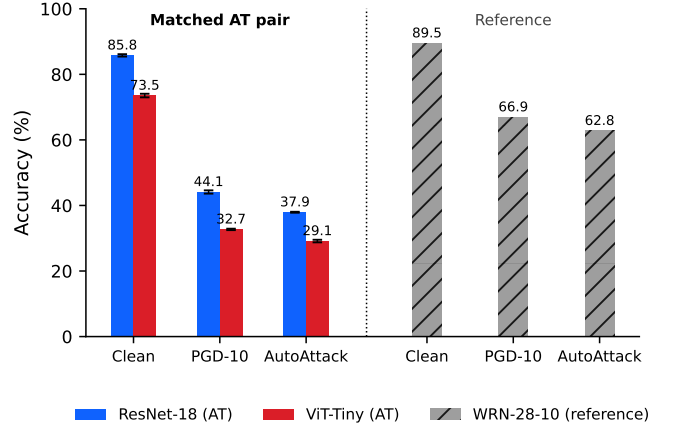


Fig. 1: Clean, FGSM, PGD-10, and AutoAttack accuracy at $\epsilon = 8/255$ on the full CIFAR-10 test set. The WideResNet AutoAttack bar is the value reported by RobustBench.

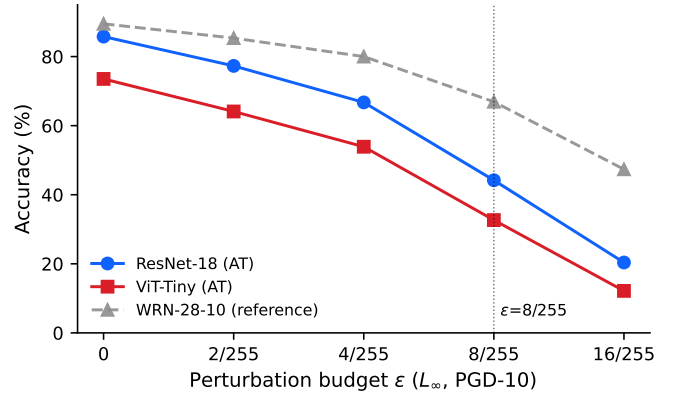


Fig. 2: PGD-10 robust accuracy vs. perturbation budget (ϵ), measured on the full test set. Both AT architectures degrade with similar shapes as epsilon increases, with the CNN maintaining a consistent margin.

defensible conclusion is narrower and more useful than a mechanism: an attribution of this kind should not be drawn from a single training run, and a paper that reports only the absolute score hides which factor is doing the work.

Figure 1 visualizes this robustness gap across architectures. To further understand the effect of perturbation magnitude, Figure 2 shows PGD-10 robust accuracy across perturbation budgets $\epsilon \in \{2/255, 4/255, 8/255, 16/255\}$ on the full test set.

Figure 3 provides visual examples of adversarial perturbations and their effects on model predictions.

B. Transfer Attack Analysis

This section contains the paper’s central measurement. We craft PGD-10 adversarial examples on a source model and evaluate them on a target, on the full test set, for each of the three seed pairs. Because outcomes are logged per sample, the same attack run can be scored under several conditioning

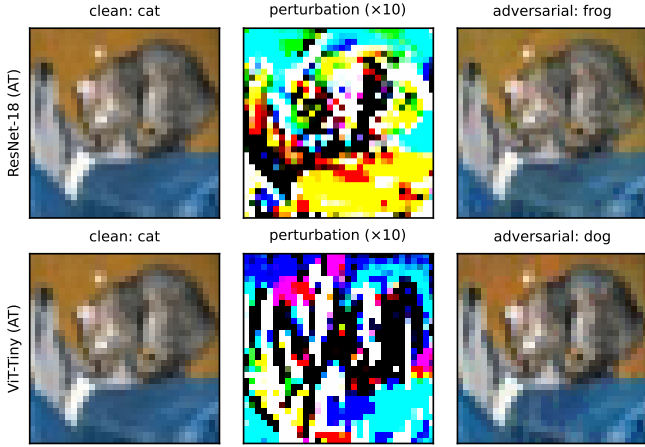


Fig. 3: White-box PGD-10 attacks on each model and a CNN $\rightarrow$ ViT transfer attack, each on a different test image, showing the clean prediction, the perturbation magnified $10\times$, and the adversarial prediction. Perturbations are bounded by $\epsilon=8/255$ and remain visually subtle at this scale.

TABLE III: Transfer asymmetry under the unconditioned rate and three established conditioning protocols (PGD-10, $\epsilon=8/255$, full test set; mean $\pm$ std over three seed pairs). The rightmost column repeats the difference measured on our earlier single-run checkpoints.

Protocol	CNN $\rightarrow$ ViT	ViT $\rightarrow$ CNN	Diff.	Diff. [‡]
Unconditioned (raw)	41.02 $\pm$ 0.55	27.45 $\pm$ 0.27	+13.57 $\pm$ 0.33	+8.27
Target-correct	19.87 $\pm$ 0.18	15.51 $\pm$ 0.59	+4.36 $\pm$ 0.44	+0.63
Both-correct	18.25 $\pm$ 0.17	9.98 $\pm$ 0.31	+8.27 $\pm$ 0.23	+5.33
Successful-source	36.39 $\pm$ 0.76	17.02 $\pm$ 0.52	+19.37 $\pm$ 1.27	+11.17

[‡]Single run, validation split seen during clean pre-training.

protocols without recomputation: the protocols differ only in which samples enter the denominator (Section III-F).

The denominators differ sharply between protocols. For CNN $\rightarrow$ ViT and ViT $\rightarrow$ CNN respectively they are 10,000/10,000 without conditioning, 7,353/8,579 under target-correct, 7,061 shared samples under both-correct, and 2,831/3,814 under successful-source. The spread quoted in the Diff. column is the standard deviation of the paired within-seed difference, which cannot be recovered from the two per-direction spreads because the directions covary across seeds, and it is the training-run reference used in Section IV-I. The result is that the protocol, not the model, dominates the measured quantity. On identical models and identical data, the asymmetry ranges from $+4.36\pm0.44$ to $+19.37\pm1.27$ points; the mean per-seed spread between the largest and the smallest protocol estimate is 15.01 ± 0.84 points, a factor of 4.4, whereas the seed-level standard deviation within any single protocol stays below 1.5 points. The protocol also decides the statistical verdict. Under the both-correct protocol, which scores both directions on the same samples and therefore supports paired inference, the difference is 8.27 ± 0.24 points, differing in the third decimal from the ±0.23 of the table only because the two estimators differ, with a paired bootstrap CI of [7.33, 9.22], a sign-flip permutation $p < 5 \times 10^{-5}$ in every

seed, and TOST equivalence rejected at all three margins (±1 , ±2 and ±3 points, $p = 1.0$ in each); under the target-correct protocol on our earlier checkpoints, the same pipeline accepted equivalence at ±2 points. A reader given only one of these numbers would draw a different conclusion about whether the architectures differ at all.

A protocol name is not by itself a specification. The successful-source protocol conditions on samples whose attack succeeded on the source, and whether that phrase requires the source to classify the clean input correctly is a sub-choice that transfer studies rarely state. We report the stricter reading, in which white-box success means that the source is correct on the clean input and wrong on its adversarial counterpart. Under the looser reading, which also admits samples the source already misclassified, the measured asymmetry falls from $+19.37$ to $+14.60$ points and the protocol spread from 15.01 to 10.45 points. That shift of 4.77 points is larger than the run-to-run standard deviation of any protocol in Table III by more than a factor of three, and the other three protocols are unchanged by construction. Because a sub-choice of this size is invisible in a protocol’s name, all four conditioning masks are defined in a single place in the released code.

What is stable is the direction. CNN-crafted examples transfer better to the ViT than the reverse under all four protocols, in all three seeds, and also under the momentum-based MI-FGSM [41] (19.07 ± 0.33 versus $15.42 \pm 0.52\%$ target-correct), so the effect is not an artifact of the PGD attack itself. The magnitude, its significance, and any equivalence claim are protocol-dependent; the sign is not.

The mechanism behind the spread is compositional. Samples that the target solves on clean input but the source does not are three to four times more fragile than jointly solved ones ($41.2 \pm 0.8\%$ and $59.1 \pm 1.7\%$ fooling versus 10.0 and 18.3%), and these marginal subsets differ five-fold in size ($1,517 \pm 102$ versus 291 ± 16) because the two models differ in clean accuracy. Conditioning on target-correct admits both marginal subsets, whose asymmetric sizes pull the two rates toward each other; conditioning on both-correct excludes them, and the directional difference reappears. This account covers three of the four protocols. The successful-source protocol is driven differently, and separating the two drivers changes what the spread depends on: across 18 direction pairs drawn from six seed-and-dataset clusters, the spread of the four-protocol asymmetry *falls* as the clean-error gap between the two models widens (slope -0.528 , cluster-bootstrap CI $[-0.664, -0.418]$), whereas the spread computed without the successful-source protocol *rises* with it (slope $+0.431$, CI $[+0.333, +0.633]$). The successful-source rate is the smallest of the four in 12 of 18 pairs and the largest in 4 more, so it is the extreme in 16. The protocol pair with the largest average gap is target-correct against successful-source (23.77 points), and it is the widest pair in 13 of the 18 directions; in the remaining five, all on CIFAR-100, the unconditioned rate is the far end instead. Conditioning on source success selects by the *source*’s own robustness, which is not a function of the clean-accuracy gap between the pair, so it contributes a second and partly opposing term. The pool for this fit is the one fixed in the pre-registration, which assigns the SVHN pair to an independent

consistency check rather than to the fitted points. Adding its two directions, whose clean-error gap of under two points falls in an otherwise unoccupied band of the x axis, moves the four-protocol slope to -0.133 with a cluster-bootstrap CI of $[-0.570, +0.469]$ that includes zero, while the three-protocol slope keeps both its sign and its interval. We report the sensitivity rather than absorb it, because it shows how thinly the four-protocol slope is supported.

The two arms agree on three protocols. We calibrate the protocol spread against the clean-error gap in two ways that fail differently. The *controlled* arm holds one member of a pair fixed and sweeps the other along its own training trajectory, so architecture, seed, dataset, and attack budget stay constant while the gap moves (116 points from eight seed-and-dataset clusters). The *observational* arm compares final models across architectures, seeds, and datasets, where everything moves at once (18 direction pairs from six clusters). Inference in both is by cluster bootstrap at the trajectory level; the degrees of freedom are set by the number of clusters, not by the number of points, and we do not pool the arms. Two cautions apply to these intervals. With six to eight clusters a percentile cluster bootstrap undercovers, so the intervals below should be read as lower bounds on width rather than as exact. And the WideResNet is a single fixed external checkpoint that appears in all six observational clusters, in twelve of the eighteen direction pairs, so those clusters are not fully independent; the pair-only sensitivity that excludes it is reported alongside.

Excluding the successful-source protocol, the two arms give slopes of $+0.673$ (CI $[+0.602, +0.727]$) and $+0.431$ (CI $[+0.333, +0.633]$): same sign, overlapping intervals. Including it, the controlled arm gives $+0.273$ (CI $[+0.219, +0.371]$) while the observational arm gives -0.528 (CI $[-0.664, -0.418]$). Composition accounts for most of that disagreement without closing it. The controlled arm contains no WideResNet, and restricting the observational arm to the same architecture pair moves its four-protocol slope from -0.528 to -0.100 (CI $[-0.464, +0.092]$), an interval that includes zero but still does not overlap the controlled arm's. The two arms therefore agree on the three-protocol slope and do not agree on the four-protocol one, which is the ordering the mechanism claim rests on. Two limits belong with this: the observational arm's negative slope is fitted across an unmeasured gap of 7.2 points on the x axis, so it compares two separated clusters rather than tracing a trend; and the controlled arm's slope survives dropping early checkpoints ($+0.201$ to $+0.492$ across thresholds), with the three-protocol slope the most stable quantity of all (0.661 to 0.673).

1) *How much of the raw rate is the target's clean error?:* To quantify the confound directly rather than argue it, we extend the analysis to a 3×3 matrix by adding the WideResNet-28-10 reference as both source and target (Table IV). The relationship between the two rates turns out to be stronger than a correlation: it is an identity resting on a single empirical fact.

Write e for the target's clean error, r_{raw} for the unconditioned rate and r_{cond} for the target-correct conditioned rate. If a sample the target already misclassifies on clean input is still misclassified after the perturbation, then $r_{\text{raw}} = e + r_{\text{cond}}(1 - e)$,

and therefore

$$r_{\text{raw}} - r_{\text{cond}} = e(1 - r_{\text{cond}}). \quad (11)$$

The antecedent is the part that is measured, and it holds tightly: across 36 off-diagonal directions (two datasets, three seed pairs, three targets, two sources) the probability that a clean-misclassified sample is also misclassified under attack ranges from 0.989 to 1.000. Equation (11) consequently reproduces every measured unconditioned rate to within 0.41 points, with a mean absolute residual of 0.095 points; the residual accounts for at most 1.39% of the deviation being explained.

This changes the status of the claim, and strengthens it. The deviation of unconditioned from conditioned transfer rates does not merely correlate with the target's clean error in our data; it is that error, scaled by $(1 - r_{\text{cond}})$, up to a residual below half a point. Unconditioned transfer rates are contaminated by target weakness *by construction*, in any study whose targets satisfy the antecedent, not as a regularity we happened to observe on three targets. The antecedent is itself a measurable quantity and should be measured rather than assumed. Writing p for the probability that a clean-misclassified sample stays misclassified under attack, the general form is $r_{\text{raw}} = e(1 - p) + r_{\text{cond}}(1 - e)$, which reduces to Equation (11) when $p = 1$. We measure p between 0.989 and 1.000 on adversarially trained targets, where the correction is therefore exact to within a few tenths of a point. On standard-trained targets, which is the regime most transfer studies attack, we observe p as low as 0.88 in our own earlier runs, and there the identity becomes a bound rather than an equality. We previously stated this result as a Pearson correlation ($r = 0.997$ over six directions drawn from three targets) and that phrasing understated it: a near-unit correlation between a quantity and its own dominant component is arithmetic, and the small-sample caveat it required is unnecessary once the identity is written down.

What remains dataset-specific is the proportionality factor, not the mechanism. Because $(1 - r_{\text{cond}})$ depends on how transferable the attack is, the fitted slope is 0.762 points per point of target error on CIFAR-10 and 0.656 on CIFAR-100. It is an average over the targets measured rather than a constant, and it should be re-estimated, not assumed, when the identity is applied to other model pools. Unconditioned transfer rates therefore do not measure transferability across architectures; they measure transferability plus the target's clean weakness, in a proportion that Equation (11) makes explicit.

Figure 4 shows the conditioned matrix. The third architecture also reframes the asymmetry itself. Averaged over the two foreign sources, incoming conditioned transfer is 18.50% for the ResNet target, 19.04% for the ViT target, and 10.57% for the WideResNet target, which tracks each target's own white-box conditioned fooling rate (48.52, 55.57 and 25.16%) with $r = 0.986$ over the three targets. On this evidence the CNN→ViT advantage is better read as a property of the weaker target than as evidence that CNN-crafted perturbations are intrinsically more transferable: the most robust target absorbs attacks from both families almost equally well. With

TABLE IV: 3×3 transfer matrix (PGD-10, $\epsilon=8/255$, full test set; mean $\pm$ std over three seed pairs; the WideResNet column and row use the single external checkpoint). Raw = unconditioned fooling rate, Cond. = target-correct conditioned rate. Diagonal entries are white-box attacks.

Source	Target	Raw (%)	Cond. (%)
ResNet-18 AT	ResNet-18 AT	55.84 $\pm$ 0.53	48.52 $\pm$ 0.84
ResNet-18 AT	ViT-Tiny AT	41.07 $\pm$ 0.58	19.93 $\pm$ 0.20
ResNet-18 AT	WRN-28-10	21.69 $\pm$ 0.09	12.48 $\pm$ 0.10
ViT-Tiny AT	ResNet-18 AT	27.45 $\pm$ 0.34	15.52 $\pm$ 0.64
ViT-Tiny AT	ViT-Tiny AT	67.34 $\pm$ 0.18	55.57 $\pm$ 0.44
ViT-Tiny AT	WRN-28-10	18.25 $\pm$ 0.12	8.65 $\pm$ 0.14
WRN-28-10	ResNet-18 AT	32.63 $\pm$ 0.14	21.48 $\pm$ 0.31
WRN-28-10	ViT-Tiny AT	39.73 $\pm$ 0.25	18.15 $\pm$ 0.28
WRN-28-10	WRN-28-10	33.03	25.16

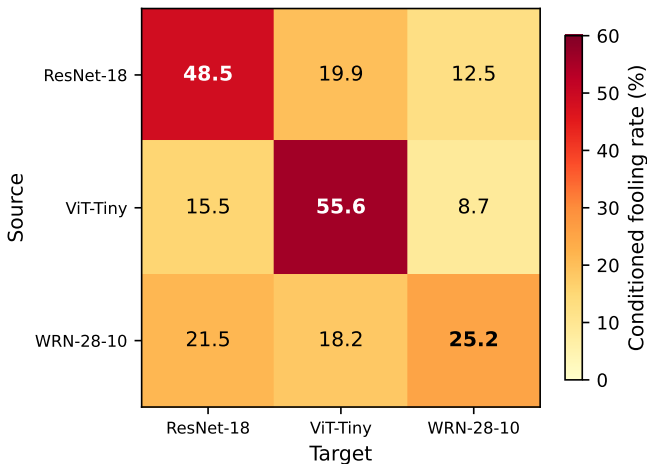


Fig. 4: Conditioned (target-correct) transfer fooling rates, mean over three seed pairs. Off-diagonal asymmetry under this protocol is +4.4 points; Table III shows how far the same quantity moves under other protocols.

only three targets this correlation is suggestive rather than conclusive, and we report it as such.

C. Does the Protocol Effect Generalize? A Second Dataset

The measurement above is made on one dataset, and a protocol effect that exists only on CIFAR-10 would not support a general reporting recommendation. We therefore repeated the whole design on CIFAR-100: same architectures, same threat model, same attack budget, same three-seed structure, same four protocols. Before running it we registered three directional predictions and three endpoint bands in an appendix-only document, so that the outcome could not be reinterpreted after the fact.

Across the same four protocols the measured CNN $\rightarrow$ ViT asymmetry runs from $+4.96 \pm 1.01$ points (target-correct) to $+18.53 \pm 0.71$ points (unconditioned), with $+10.92 \pm 0.34$ under both-correct and $+17.50 \pm 0.92$ under successful-source. The mean per-seed spread is 13.83 ± 1.30 points, against 15.01 ± 0.84 on CIFAR-10, so the harder dataset places the protocol effect in the same range rather than shrinking it. The direction is positive in all twelve measurements (four

TABLE V: CIFAR-100 replication (mean $\pm$ std over three independent training runs per architecture, full test set, $\epsilon=8/255$). Both models are adversarially trained with the protocol of Section III-E. Absolute robustness is lower than on CIFAR-10 for both architectures, and the CNN-over-ViT ordering is unchanged.

Model	Clean (%)	PGD-10 (%)	AutoAttack (%)
ResNet-18 AT	63.86 $\pm$ 1.05	19.30 $\pm$ 0.32	15.04 $\pm$ 0.54
ViT-Tiny AT	43.17 $\pm$ 1.10	11.15 $\pm$ 0.60	8.87 $\pm$ 0.83

protocols $\times$ three seed pairs), as it was on CIFAR-10. Under the both-correct protocol the paired difference is 10.92 points with a bootstrap CI of [9.47, 12.35], a sign-flip permutation p below the resolution of 20,000 permutations in every seed, and equivalence rejected at every margin tested.

One of the three registered predictions is confirmed, one fails under the reading we treat as binding, and the third is not testable in this design. The prediction that the sign would be preserved is confirmed (12/12). The prediction that the protocol spread would be larger on the harder dataset was registered as a comparison against the CIFAR-10 spread, quoted at the time as 10.45 points. Against that literal threshold it holds ($13.83 > 10.45$); against the CIFAR-10 spread as it now stands under the successful-source definition of Section IV-B it does not ($13.83 < 15.01$). We report the failure rather than the reading that would rescue it, because the registration named the CIFAR-10 spread and not the number. The two datasets place the protocol effect in the same range instead of ordering it by difficulty, so the mechanism we propose predicts the size of the effect but not its ordering between these two datasets. The third predicted that the deviation of the unconditioned from the conditioned rate would grow with the target’s clean error. The measured slope is $+0.656$ points per point of target error, positive as predicted; but on CIFAR-100 two of the three targets have clean errors that differ by 0.21 points (36.15 and 36.36), so the correlation is computed over two effective clusters rather than three, and its value ($r = 0.931$ over six directions, 0.974 aggregated to targets) rests on a two-point contrast. The measurement is *consistent with* the prediction and does not *test* it; we report it as such rather than as a third confirmation.

One registered endpoint was missed. We predicted clean accuracy in the 33 to 40% band for the adversarially trained ViT and measured $43.17 \pm 1.10\%$. The band was derived by assuming that the clean-accuracy cost of adversarial training scales the same way for both architectures. It does not: the ViT retained 0.885 of its clean pre-training accuracy under adversarial training, against 0.802 for the ResNet. The two adversarial endpoints, which are the ones the protocol claim depends on, fell inside their bands (PGD-10 predicted 10 to 14%, measured 11.15; AutoAttack predicted 8 to 11%, measured 8.87).

Class composition does not manufacture the asymmetry. Because conditioning changes which samples enter the denominator, it also changes the class mix, and an asymmetry could in principle be a composition artifact. We decompose

TABLE VI: SVHN transfer asymmetry under the four protocols (PGD-10, $\epsilon=8/255$, full test set $n=26,032$; mean $\pm$ std over two seed pairs; two architectures only, so no third-architecture control). Unlike CIFAR-10 and CIFAR-100, the sign is not the same in all four rows.

Protocol	CNN $\rightarrow$ ViT	ViT $\rightarrow$ CNN	Diff.
Unconditioned (raw)	31.89	31.49	+0.39 $\pm$ 0.08
Target-correct	26.32	27.33	-1.00 $\pm$ 0.22
Both-correct	25.36	25.70	-0.35 $\pm$ 0.01
Successful-source	61.38	58.69	+2.70 $\pm$ 0.40

each direction’s rate into a composition component and a rate component. Under target-correct the composition effect is -1.249 , -1.330 and -1.065 points across the three seed pairs, that is 12.9 to 18.6% of the asymmetry and *negative*: rebalancing the class mix would make the asymmetry larger, not smaller. Under successful-source it is small and positive in every seed ($+0.527$, $+0.435$ and $+0.382$ points, that is 2.1 to 3.1% of the asymmetry). Under the unconditioned and both-correct protocols it is exactly zero, which is the arithmetic check the decomposition must pass.

Two qualifications belong with this result. First, the two adversarial bands held with margins of 0.22 points (AutoAttack) and 0.48 points (PGD-10) at the nearest seed-level edge. Those margins are smaller than the checkpoint-selection noise on this same dataset: sweeping early-stopping patience and curve smoothing over the CIFAR-100 trajectories, with the training run held fixed, moves the reported PGD-10 accuracy by 0.18 to 1.83 points (mean 0.86). The same nine-cell sweep on CIFAR-10 gives 0.54 to 2.83 points per validation split, and the eighteen-cell figure quoted in Section IV-I is larger still because it also varies the split, a dimension the CIFAR-100 protocol does not have. The bands held, but by margins at or below the selection noise of the dataset they were registered for; a reader should weigh the pre-registration accordingly. Second, the same held-out split serves selection at both stages rather than two independent splits. That choice is inherited deliberately from the CIFAR-10 protocol so the two datasets remain comparable, and Section IV-I reports what it does and does not cost.

D. A Third Dataset Where the Architectures Are Matched

CIFAR-10 and CIFAR-100 both place the two architectures far apart in clean accuracy (about 11 and 21 points). SVHN [62] does not (Table V reports the CIFAR-100 absolutes for comparison): adversarially trained under the same recipe, the ResNet reaches $94.27 \pm 0.70\%$ clean and the ViT $92.42 \pm 0.56\%$, a gap of 1.85 points. Robust accuracies are 55.52 ± 0.64 and 52.30 ± 0.10 under PGD-10. This is the regime our thesis makes the sharpest prediction about, and the prediction is not the one a reader of the previous sections would expect.

When the architectures are matched, the protocol decides the sign. Two of the four protocols report a CNN advantage and two report a ViT advantage (Table VI). The split is not arbitrary. The two protocols we endorse for directional claims, target-correct and both-correct, agree with each other

and both favor the ViT; the sign change separates them from the unconditioned rate and the successful-source rate. The unconditioned reversal is moreover the arithmetic consequence of Equation (11): the ViT target’s clean error exceeds the CNN’s by about two points here, which is enough to carry a -1.00 conditioned difference to $+0.39$ raw. Reporting a raw rate can therefore flip the sign of a published comparison, and the identity says in advance when it will. Under the both-correct protocol, which we treat as primary because it scores both directions on identical samples, the difference is -0.35 points with a paired bootstrap CI of $[-0.77, +0.06]$, a sign-flip permutation p of 0.105, and TOST equivalence *accepted* at all three margins, including the tightest one we test (± 1 point, $p = 0.0007$). A study reporting only the successful-source rate would announce a 2.70-point CNN advantage on SVHN; one reporting only the target-correct rate would announce a 1.00-point ViT advantage. Both would be arithmetically correct and neither would be reproducible by the other.

This behaviour is consistent with the mechanism rather than an exception to it. The protocol spread on SVHN is 3.70 ± 0.62 points, against 15.01 ± 0.84 on CIFAR-10 and 13.83 ± 1.30 on CIFAR-100, an order of magnitude apart from the two CIFAR values while the clean-accuracy gaps are 1.85, ≈ 11 and ≈ 21 points. The relation is not monotone across the three: the largest gap does not produce the largest spread, and CIFAR-100 sits slightly below CIFAR-10. What separates the datasets here is the near-matched pair, not the ordering among the mismatched ones. Once the spread is small enough it no longer clears the asymmetry itself, so the sign becomes a property of the protocol rather than of the architectures. The direction we reported as stable in Section IV-B is stable *in the regime where the two models differ in clean accuracy*, which is the regime both CIFAR datasets happen to occupy.

SVHN was run in a reduced configuration fixed in advance: two seed pairs rather than three, 50 adversarial-training epochs, no AutoAttack, and no third architecture. The last of these means the 3×3 confound control of Section IV-B1 cannot be built here, and we do not build a within-SVHN version of it.

E. Does the Protocol Effect Survive a Change of Threat Model?

Everything so far is measured inside one threat model. To test whether the protocol effect is an L_∞ artifact, we re-evaluated the same CIFAR-10 checkpoints under an L_2 budget of $\epsilon = 0.5$, registering three directional predictions before looking at any L_2 output. The models are L_∞ -trained, so this measures the *measurement axis*, not L_2 robustness, and the absolute values below are not comparable with L_2 -trained references.

All three registered predictions hold. The protocol spread is 10.92 points under L_2 against 15.01 under L_∞ , so it is smaller under the changed norm but stays far above the two-point floor the pre-registration set and an order of magnitude above the run-to-run standard deviation. The sign is positive in all twelve measurements, and the deviation of unconditioned from conditioned rates still rises with the target’s clean error (slope $+0.914$; we report the slope rather than a correlation

TABLE VII: Same checkpoints, L_2 budget ($\epsilon=0.5$; mean $\pm$ std over three seed pairs). PGD-L2 uses the full test set; AutoAttack-L2 uses a pre-registered 5,000-sample subset, so its clean column differs slightly from the full-set value and the two are not drawn from the same samples.

Model	Clean	PGD-L2	Clean (5k)	AA-L2
ResNet-18 AT	85.78	63.78 $\pm$ 0.38	85.33	60.61 $\pm$ 0.42
ViT-Tiny AT	73.53	51.72 $\pm$ 0.23	73.61	49.85 $\pm$ 0.20

because the eighteen directions take only seven distinct target-error values, six of them the same fixed WideResNet, so a correlation over them would overstate the precision in the way Section IV-B describes). The four protocols (Table VII) report asymmetries of $+13.03 \pm 0.81$ (unconditioned), $+2.12 \pm 0.18$ (target-correct), $+4.07 \pm 0.36$ (both-correct) and $+12.06 \pm 0.23$ (successful-source). Absolute robustness keeps the CNN-over-ViT ordering under both attacks, with McNemar $p < 10^{-57}$ in every seed pair.

The slope also confirms the identity out of sample. Equation (11) predicts a slope of $1 - r_{\text{cond}}$ averaged over targets. Conditioned rates are far lower under this L_2 budget than under L_∞ (roughly 2 to 9% against 9 to 22%), so the identity predicts a *steeper* slope, and the measured 0.914 sits where it should, against 0.762 on the same models under L_∞ . The proportionality factor is a property of the attack budget, not a constant, exactly as Section IV-B1 states.

F. Gradient Characteristics

To characterize architectural differences in input sensitivity, we analyze the gradient properties of both architectures, together with a CIFAR-native ViT control (32×32 inputs, 4×4 patches, no upsampling) trained with the same adversarial objective but before the fixed-split protocol described in Section III-E was adopted, which isolates the effect of the $32 \rightarrow 224$ upsampling in our main ViT pipeline. Table VIII summarizes the key statistics using scale-invariant sparsity measures (Section III-F); gradients are computed with per-sample losses so that norms do not depend on the evaluation batch size. Alignment there is the mean absolute cosine similarity over all $500 \times 499/2$ sample pairs, and every ResNet to ViT difference on the scale-invariant sparsity metrics is significant under per-sample paired Wilcoxon tests in every run, with Holm-corrected $p < 10^{-11}$ and paired Cohen’s d between 0.32 and 1.23.

Across all three scale-invariant sparsity measures, CNN gradients are sparser than ViT gradients, and the ordering reproduces in every run with seed-level standard deviations at least three times smaller than the gap. Because the same 500 samples are measured under both models, the differences are tested with per-sample paired statistics: all three are significant in all three runs (Wilcoxon signed-rank, Holm-corrected $p < 10^{-11}$ throughout) with paired Cohen’s d between 0.32 and 1.23. The ordering is a product of adversarial training rather than of architecture: in the clean-trained counterparts of the same runs the ViT is the sparser model (Hoyer 0.380 ± 0.009 versus 0.347 ± 0.002), so adversarial training sparsifies CNN

TABLE VIII: Gradient characteristics computed from per-sample loss gradients ($n=500$ test samples per run; mean $\pm$ std over three runs).

Metric	ResNet-18 AT	ViT-Tiny AT	ViT (native) [‡]
Hoyer sparsity	0.4928 $\pm$ 0.0120	0.4561 $\pm$ 0.0055	0.405
Gini coefficient	0.6498 $\pm$ 0.0089	0.6099 $\pm$ 0.0051	0.571
Frac. <1% of max	34.8% $\pm$ 2.1	26.8% $\pm$ 0.9	17.9%
Alignment (absolute)	0.0378 $\pm$ 0.0009	0.0562 $\pm$ 0.0004	0.079
Alignment (signed)	0.0014	0.0005	–

[‡]CIFAR-native ViT control (4×4 patches, no upsampling; 5.4M params), a single checkpoint predating the fixed-split protocol, selected on the test set, and included only for the sign of the upsampling effect, not as a quantitative comparison.

gradients far more strongly, consistent with [28]. The native-resolution ViT control is less sparse than the upsampled ViT on every measure, so the ordering is not an artifact of the $32 \rightarrow 224$ upsampling. Cross-sample alignment is higher for the ViT in absolute cosine, and the clean-trained pair already shows this gap (0.058 versus 0.029), which makes it a property of the architecture rather than of the defense. The *signed* mean cosine is near zero for both models (≤ 0.0014); what samples share is sensitivity axes, not consistent directions, so the alignment gap is not evidence for cheap universal perturbations.

These gradient results are also the one part of our analysis that is insensitive to which checkpoint set is used: every quantity in Table VIII reproduces the value measured on our earlier single-run checkpoints to within the reported standard deviations, in contrast to the conditional decomposition of Section IV-A.

1) *Sparsity is not spatial locality*: Sparsity measures answer how *many* gradient components are large, not whether those components sit together in the image. The distinction matters because the sparser-CNN result invites a spatial reading, in which gradients are taken to be more “localized” or “concentrated” on object regions. We do not attribute that reading to a particular study; we tested it because it is the natural interpretation of a sparsity result and because our own earlier figures encouraged it. The measures do not support it. We therefore measure spatial structure directly on the same gradients, using four scale-invariant descriptors of the channel-summed energy map: the smallest pixel fraction carrying 50% and 90% of the gradient energy, the spatial entropy of the normalized energy map, and Moran’s I with 4-neighbourhood weights, which is positive when energy clusters in adjacent pixels.

The result is negative (Table IX). None of the four measures shows a significant difference at the run level. The largest of them, the area carrying 90% of the energy, points toward a *smaller* area for the CNN by two percentage points of image area and does not survive correction, which is a marginal effect next to a Hoyer gap that is significant at $p < 10^{-11}$. Per-sample tests within a single run reach much smaller p -values on these measures, but they answer a different question, whether two fixed checkpoints differ on these 500 images, and the sign is not stable across runs on two of the four measures. Neighbourhood clustering is essentially identical for the two architectures. The two families of measures therefore

TABLE IX: Spatial locality of input gradients ($n=500$ samples per run; mean $\pm$ std over three runs). Paired differences are computed on the same samples. Inference is at the run level: p is a one-sample t -test on the three per-run mean differences, Holm-corrected across the four measures, so that the unit of inference matches the unit of the claim. Lower area and entropy mean more localized; higher Moran’s I means more clustered.

Measure	ResNet-18	ViT-Tiny	Diff.	p
Area for 50% energy	0.0378 $\pm$ 0.0017	0.0397 $\pm$ 0.0010	−0.0019	0.54
Area for 90% energy	0.2345 $\pm$ 0.0078	0.2570 $\pm$ 0.0067	−0.0224	0.40
Spatial entropy (nat)	5.2014 $\pm$ 0.0387	5.2524 $\pm$ 0.0197	−0.0509	0.54
Moran’s I	0.4141 $\pm$ 0.0132	0.3941 $\pm$ 0.0024	+0.0200	0.47

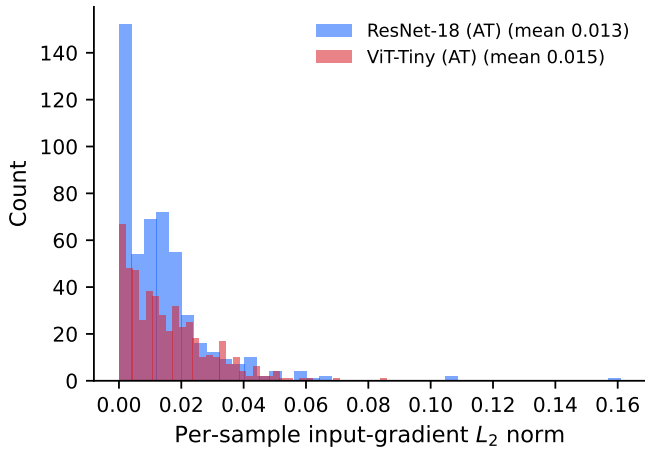


Fig. 5: Distribution of per-sample gradient L_2 norms for both architectures. The magnitude maps for the same inputs are shown in Figure 6.

disagree, and the disagreement is informative: adversarially trained CNN gradients place more of their mass on fewer pixels, but those pixels are not more spatially grouped than the ViT’s. Descriptions of CNN gradients as spatially localized are not supported by our measurements, and we restrict our own claims to sparsity.

Figure 5 provides visual evidence for these gradient differences.

Figure 6 shows a direct comparison of gradient heatmaps, highlighting the structural differences in how each architecture computes input gradients.

G. Feature Degradation in Vision Transformers

We next measure where adversarial damage accumulates along depth, extending layer-wise representation analyses of clean-trained ViTs [29] to the adversarially trained setting. Two design choices in this measurement are usually left implicit and turn out to matter, so we vary both: the sample size ($n = 1000$ per run rather than a single small batch) and the token aggregation used to summarize a transformer block. For the ViT we report the CLS token alone, the mean over the 196 patch tokens, and the flattened concatenation of all

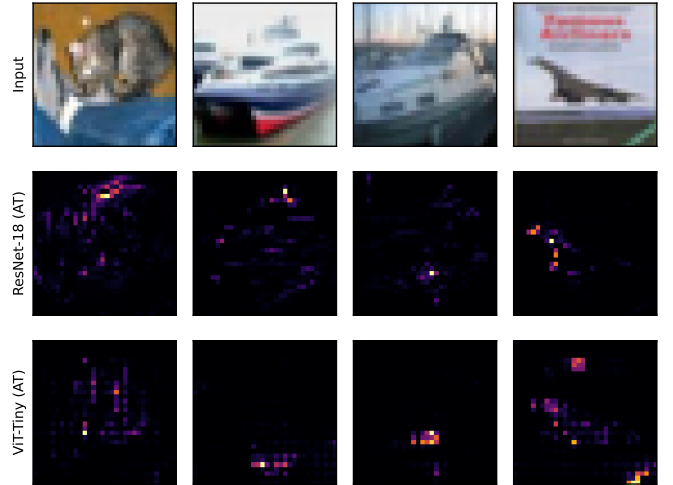


Fig. 6: Input-gradient magnitude maps for the same test images under both models. CNN gradients place more of their mass on fewer components (Table VIII); Table IX shows that this is not accompanied by a difference in spatial arrangement, so the panels should be read as sparsity rather than as localization.

TABLE X: Block-wise clean-to-adversarial cosine similarity under each model’s own white-box PGD-10 attack ($n=1000$ per run; mean $\pm$ std over three runs). ViT columns differ only in how a block’s tokens are aggregated.

Blk	CLS	ViT-Tiny AT		ResNet-18 AT	
		Patch mean	All tokens	Block	Cosine
1	0.9964	0.9990	0.9975	layer1.0	0.9951
2	0.9893	0.9954	0.9911	layer1.1	0.9918
3	0.9785	0.9928	0.9821	layer2.0	0.9879
4	0.9709	0.9877	0.9734	layer2.1	0.9853
5	0.9652	0.9870	0.9719	layer3.0	0.9753
6	0.9596	0.9873	0.9712	layer3.1	0.9607
7	0.9497	0.9870	0.9698	layer4.0	0.8777
8	0.9414	0.9840	0.9663	layer4.1	0.9108
9	0.9353	0.9850	0.9675		
10	0.9343	0.9877	0.9696		
11	0.9352	0.9912	0.9753		
12	0.9779	0.9955	0.9884		

tokens; for the CNN each residual block output is flattened over its spatial dimensions.

Two results follow. First, the two architectures differ less in *whether* damage accumulates than in *where*. In the ViT the drift is gradual and shallow: similarity declines steadily to a minimum around blocks 8 to 10 and then recovers toward the output, while feature norms stay within 0.7% of their clean values throughout, so the perturbation rotates the representation rather than rescaling it. In the CNN the damage is concentrated: the first three stages lose little, then the first block of the last residual stage collapses to 0.8777 ± 0.0181 together with a $-13.14 \pm 1.89\%$ norm contraction, after which the final block partially repairs the representation (0.9108 ± 0.0060 , norm $+5.84 \pm 3.53\%$). Adversarial damage in the CNN is thus a late, localized, magnitude-changing event; in the ViT it is a distributed, direction-only rotation. We note that on our earlier checkpoints the CNN profile fell monotonically to the last block, with no recovery, so the shape of the CNN curve,

unlike the ViT profile, should be read as recipe-dependent.

Second, the aggregation choice changes the measured profile substantially. At their respective minima the CLS pathway drifts about four times as far as the patch-token mean (0.9343 versus 0.9840, a ratio of 4.1 in $1 - \cos$), and the two summaries even place the minimum at different depths (block 10 versus block 8). Since the CLS token is what the classifier reads, this is not a nuisance but a result: the perturbation concentrates on the classification pathway while the average patch representation is comparatively preserved. It also means that layer-wise drift curves from different papers are not directly comparable unless the aggregation is stated.

Table X reports the block-wise profiles. A third measurement is negative. Adversarial perturbation does not measurably change the entropy of the CLS attention distribution: over $n = 1000$ samples per run, the mean per-layer change is at most 0.005 nats in absolute value against layer means near 4.9 nats, with a seed-level standard deviation of the same order. We state this as equivalence rather than as a failure to reject: with a margin of ± 0.05 nats, one percent of the layer mean and fixed from the scale of the measure rather than from the data, TOST accepts equivalence in all twelve layers, the weakest at $p = 0.006$. What does grow with depth is the *displacement* of the attention map, from 0.028 total variation in the first layer to 0.084 in the last, small in absolute terms, since total variation is bounded by 1. The attack therefore redirects attention slightly without broadening or sharpening it. We flag this explicitly because an earlier version of this analysis, computed on an 8-sample figure batch, suggested a visible attention effect that the larger measurement does not support; any claim that adversarial attack degrades attention should be checked at this sample size before being made. We raise this against our own preliminary version rather than against a specific study in the literature.

Finally, drift magnitude does not predict attack success. Splitting the samples by whether the attack flipped the prediction, the last-block cosine similarity of flipped and unflipped ViT samples is statistically indistinguishable (0.9895 versus 0.9877); for the CNN there is a small difference in the expected direction (0.9058 versus 0.9143). Feature drift is therefore a description of how the representation moves, not a sufficient statistic for whether the prediction changes.

Complementing the quantitative measurements, Figure 7 visualizes CLS-token attention maps at layers 1, 6, and 12 for a representative sample. The visualization is consistent with the displacement measurement above, since later layers move more than early ones, but we caution that single-sample attention panels of this kind are easy to over-read: the aggregate entropy change over 1,000 samples is null, so the visible differences reflect redirection, not diffusion. Attention-shift effects under patch perturbations [32] and attacks that explicitly target the attention mechanism [31] operate in a different regime from the L_∞ -bounded perturbations studied here.

Figure 8 shows the clean and adversarial CLS attention for a representative sample together with the per-layer entropy curves; the two entropy curves lie on top of each other, which is the visual form of the null result reported above.

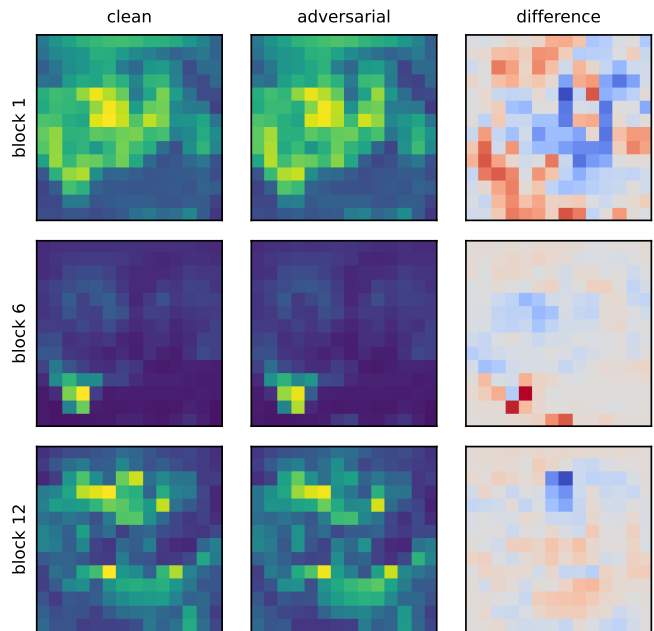


Fig. 7: CLS-token attention maps (post-softmax, head-averaged) across ViT layers under adversarial attack. Each row shows a different layer (1, 6, 12); columns show clean attention, adversarial attention, and their difference. A single sample is shown for illustration; the depth trend in attention displacement is reported quantitatively in the text over $n = 1,000$ samples per run.

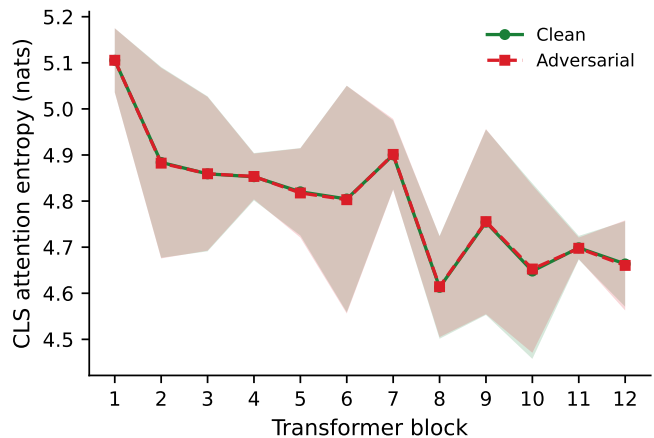


Fig. 8: Per-layer entropy of the CLS attention distribution for clean and adversarial inputs, $n = 1,000$ samples per run, mean over three runs with ± 1 s.d. shading. The attention maps themselves are shown in Figure 7.

Figure 9 shows an illustrative t-SNE embedding of the penultimate feature spaces. Because t-SNE embeddings are sensitive to hyperparameters and invite qualitative over-reading, we base all claims on metrics computed in the original feature space ($n=500$ per run, each model attacked white-box, mean $\pm$ std over three runs). The attack collapses class structure most strongly in the CNN: clean-fit 5-NN label consistency drops from $87.3 \pm 2.6\%$ to $51.3 \pm 1.8\%$, the silhouette score

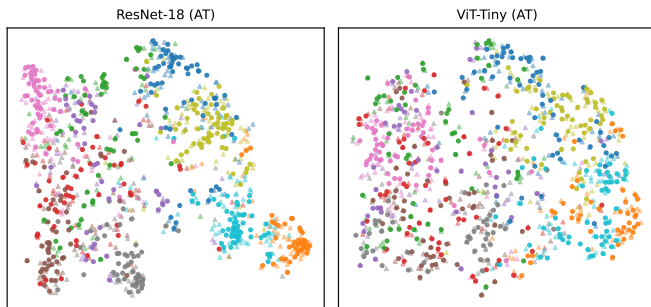


Fig. 9: Illustrative t-SNE embedding of penultimate features for clean samples (circles) and white-box PGD-10 adversarial samples (triangles), for ResNet-18 (left) and ViT-Tiny (right). t-SNE is shown for orientation only; the quantitative comparison is given in the text.

TABLE XI: ViT→CNN transfer under TGR [12] and plain MI-FGSM at a matched budget ($n=10,000$ per run; mean $\pm$ std over three runs).

Measure (%)	TGR	MI-FGSM
White-box on source (raw)	51.56 $\pm$ 0.80	66.43 $\pm$ 0.29
Transfer, target-correct	12.70 $\pm$ 0.41	15.71 $\pm$ 0.60
Transfer, both-correct	7.93 $\pm$ 0.42	10.08 $\pm$ 0.30

falls from 0.196 ± 0.010 to -0.032 ± 0.004 , the intra/inter-class distance ratio rises from 0.511 ± 0.013 to 0.847 ± 0.003 , and the relative class-centroid shift is 0.238 ± 0.012 . The ViT’s penultimate space is only weakly class-separated even on clean inputs (silhouette -0.021 ± 0.005 ; 5-NN consistency $61.1 \pm 1.4\%$, dropping to $48.5 \pm 1.4\%$), and its centroids shift far less (0.083 ± 0.004). The measurable statement is therefore that the CNN’s well-separated clean feature space is heavily disrupted by white-box attack, whereas the ViT’s class structure is weaker to begin with and moves less in these aggregate terms; this is a statement about geometry, not about robustness, since the CNN nonetheless retains the higher robust accuracy.

H. Does a ViT-Specific Transfer Attack Change the Picture?

All transfer results so far use gradient attacks that treat the ViT as a generic differentiable model. A natural objection is that the ViT direction is handicapped by this choice, since attacks designed for transformer structure are reported to transfer better. We test the objection directly with Token Gradient Regularization (TGR) [12], which suppresses extreme-valued token gradients in the attention, QKV, and MLP paths during backpropagation on the source model. Our implementation is an adaptation: the original targets standardly trained ImageNet ViTs, whereas we apply the same mechanism to our CIFAR-10 ViT-Tiny wrapper and match the attack budget to the rest of the paper ($\epsilon = 8/255$, $\alpha = 2/255$, 10 steps, momentum 1.0) rather than using ImageNet defaults. As a control we generate plain MI-FGSM examples in the same run, so the comparison is paired on identical samples and identical budget.

TGR does not improve transfer in this regime; it degrades it (Table XI). The regularization behaves as designed on the

source side, trading away 14.9 points of white-box strength, but the trade does not pay off at the target: conditioned transfer falls by 3.0 points under the target-correct protocol and by 2.2 under both-correct. Paired McNemar tests on the both-correct subset favor MI-FGSM decisively in every run ($p = 6.1 \times 10^{-10}$, 1.1×10^{-27} , 7.0×10^{-34}), so this is not a seed effect.

We state the conclusion narrowly. Our setting differs from the one TGR was designed for in three respects that could each matter: the targets are adversarially trained rather than standardly trained, the data are 32×32 CIFAR-10 images upsampled inside the model rather than native ImageNet, and the source is a 5.7M-parameter ViT-Tiny. The claim we can support is that TGR’s transfer advantage does not carry over to this regime, not that the method fails in general. For the purposes of this paper the relevant consequence is that the CNN→ViT asymmetry reported in Section IV-B is not an artifact of using architecture-agnostic attacks: giving the ViT direction a transformer-specific attack makes it weaker, not stronger.

I. Sources of Variance

Four sources of variance bear on the comparisons above. Attack-initialization stochasticity is negligible: re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed 1,000-sample subset leaves clean accuracy exactly unchanged, since the loader does not shuffle, and moves robust accuracy by a standard deviation of 0.05 points for the CNN and 0.00 for the ViT. Training-run variance is larger but still small: across the three independent runs per architecture, the standard deviation for the adversarially trained pair is at most 0.55 points on clean accuracy and 0.50 on PGD-10 (Table I).

Comparing the protocol effect against those two figures directly would be an error of scale, because the protocol effect is measured on the transfer asymmetry while both figures above are measured on absolute accuracy. We therefore quantify the protocol and training-run effects on the *same* quantity, the off-diagonal asymmetry, and report both in absolute units. Within a fixed seed pair, changing only the conditioning protocol moves the asymmetry by 15.01 ± 0.84 points. This figure is the mean of the per-seed ranges, and here it coincides with the range of the protocol means because the extremal protocols are the same in all three seeds, successful-source above and target-correct below; on CIFAR-100 the two figures differ (13.83 against 13.58) because there the extremal protocol does change between seeds. Retraining both architectures from scratch moves that same asymmetry by a standard deviation of only 0.23 to 1.48 points, the value depending on which protocol it is measured under (Table III, Diff. column). A fifteen-point measurement effect against a sub-1.5-point training effect is the ordering the paper rests on.

We deliberately do not headline the ratio of these two spreads. With three runs the denominator carries two degrees of freedom, so a χ^2 interval on any single ratio is very wide: for the successful-source protocol, whose run-to-run standard deviation is the largest of the four, the 95% interval on the ratio runs from 0.8 to 9.9 and therefore includes unity. Reported

as a sensitivity rather than an estimate, the ratio spans 5.2 to 32.6 across dispersion measures and reference protocols, and equals 28.3 under the like-for-like standard-deviation comparison on the both-correct protocol that we treat as primary. Two cautions bound it further. The protocol with the largest asymmetry also carries the largest run-to-run standard deviation, so numerator and denominator are not independent. And because all three runs share the single validation split drawn before any training, this standard deviation excludes split-to-split variance, which makes the ratio an upper bound. The absolute comparison survives the width of that interval and both cautions, which is why it, and not the fold-change, carries the claim. This ordering is the quantitative form of the paper’s argument: with a protocol left unstated, a reader cannot tell a real architectural difference from a measurement choice.

A fourth source is the offline checkpoint-selection protocol, which is usually left implicit. Holding a training trajectory completely fixed and varying only how its final checkpoint is chosen, that is which validation split, what early-stopping patience, and whether the validation curve is smoothed before the argmax (eighteen combinations), moves the reported PGD-10 accuracy by 2.62 to 2.85 points per seed for the ResNet and 1.58 to 2.09 points for the ViT. The weights on disk are identical across these cells; only the rule that picks one of them differs. About half of that dispersion comes from one of the three dimensions. Smoothing the validation curve before the argmax lowers test PGD-10 accuracy monotonically in every ResNet run and in two of the three ViT runs, so it is not a neutral alternative but a systematically worse rule; restricted to the split and patience choices that studies actually vary, the dispersion falls to 1.09 times the seed-level standard deviation for the ResNet and 1.42 times for the ViT. We report both figures because the larger one measures what an undocumented preprocessing step can add, and the smaller one measures what a documented protocol still leaves open. We report this dispersion in absolute units and deliberately not as a ratio to seed-level dispersion, because that ratio depends on which of the eighteen cells is taken as the reference: swept across all eighteen it runs from 0.67 to 3.38 for the ResNet and 0.63 to 7.38 for the ViT, falling below unity for some references and inverting the comparison.

A counterweight belongs immediately next to it. On CIFAR-100 the epoch selected by our own protocol varies by 25 epochs across the three ResNet seeds, yet the resulting test PGD-10 standard deviation is 0.32 points. The selection *path* was unstable while the selected *outcome* was not. Neither observation generalizes without the other: selection choices can move a reported number by more than a training seed does, and they can also leave it nearly untouched, and which of the two happens is a property of the trajectory’s plateau rather than of the architecture. The practical consequence for reporting is the same in both cases: the selection rule is part of the result and belongs in the methods section.

Computational Details: The training runs and analyses reported here were conducted on an NVIDIA RTX 5090 (32GB VRAM); the earlier single-run checkpoints used for the comparison in Table II were trained on an RTX 5060 Ti

(16GB). All runs used PyTorch 2.6.0 with CUDA 12.8 and deterministic cuDNN settings. AutoAttack evaluation used the standard configuration (APGD-CE, APGD-T, FAB-T, Square) on the full 10,000-sample test set in seeded 1,000-sample chunks with per-sample outcome logging. The three-seed training pipeline required 21.7 GPU-hours in total.

V. DISCUSSION

A. What the Measurements Show

The central result is an ordering of dispersions. Measured on the transfer asymmetry itself, retraining both architectures from scratch moves it by a standard deviation of 0.23 to 1.48 points, whereas changing only the conditioning protocol moves it by 15.01 ± 0.84 points (Section IV-I). We state the two spreads in absolute units rather than as a fold-change, because with three runs the interval on that ratio is too wide to carry a headline. For scale on a different quantity, re-running an attack with a new random initialization moves absolute robust accuracy by about 0.05 points and retraining moves the adversarially trained pair by at most 0.55. A quantity that the measurement choice moves by fifteen points while retraining moves it by one is not a stable property of the architectures, and reporting it without the protocol communicates almost nothing.

One objection deserves an answer in its own terms. The four protocols estimate four different quantities, so the distance between them is not a variance and calling it one would be a category error. We agree, and we are not claiming that the architectures have a single true asymmetry that the protocols estimate with error. Our claim is about what a reader can recover. When a study reports one number without naming the protocol, the reader cannot tell which of the four quantities was estimated, and the spread we measure is the size of that ambiguity. It is a bound on what the omission costs, not a dispersion of a single estimator, and it is the reason the protocol belongs in the result. This is not an argument against conditioning, since conditioning on correctly classified samples is established practice [9], [40], [41] and remains necessary, but an argument that the choice belongs in the result, not in a footnote.

Nor is it an argument that our direction is the correct one. Under a both-correct protocol comparable to ours, Mahmood et al. [8] report the opposite direction on a different architecture pair, finding transfer into the transformer weaker than transfer out of it. Our mechanism is compatible with that result: we attribute the asymmetry to the target’s own vulnerability rather than to any special strength of CNN-crafted perturbations, and the weaker target in their pair is not the weaker target in ours. Testing this properly would require their per-direction clean accuracies under a shared protocol, which is exactly the reporting we ask for and the sharpest external test the mechanism admits.

The four protocols are not interchangeable, and each has a defensible use. The unconditioned rate should not be used for architecture comparison at all: our 3×3 matrix shows its deviation from the conditioned rate is algebraically equal to the target’s clean error scaled by $(1 - r_{\text{cond}})$, which means

it measures target weakness in a known proportion rather than transferability. The target-correct protocol removes that confound but admits samples the source itself cannot solve, and because those marginal subsets are hyper-vulnerable and asymmetrically sized, it pulls two directions toward each other by an amount that depends on the clean-accuracy gap. The both-correct protocol scores both directions on identical samples and is the only one of the four that supports paired inference, which is why we treat it as the primary estimate. The successful-source protocol answers a different question, how well an attack that already worked carries over, and its larger value reflects that conditioning on source success selects strong perturbations.

Across four protocols, three training runs, and two attack families (PGD-10 and MI-FGSM), adversarial examples crafted on the CNN transfer better to the ViT than the reverse *on both CIFAR datasets*. On SVHN, where adversarial training under the same recipe leaves the two architectures 1.85 points apart in clean accuracy rather than 11 or 21, that is no longer true: two protocols report a CNN advantage and two report a ViT advantage, and under the primary both-correct protocol the difference is -0.35 points with equivalence accepted (Section IV-D). The direction is a property of the unmatched regime, not of the architectures. Within the unmatched regime what changes across protocols is only how large the effect looks and whether an equivalence test accepts it; once the models are matched, the sign changes too. Our three-target analysis suggests a reading of this direction that does not require the CNN’s perturbations to be special: incoming transfer tracks each target’s own white-box vulnerability ($r = 0.986$ over three targets), and the most robust target absorbs attacks from both families almost equally. On this account the asymmetry is a statement about which model is the weaker target, which is consistent with our finding that a transformer-specific attack [12] does not rescue the ViT direction: with three targets the correlation is suggestive, and a larger model pool would be needed to establish it.

The conditional decomposition separates absolute robustness into a clean-accuracy factor and a conditional-susceptibility factor. In our corrected runs both favor the CNN, so the decomposition and the headline agree. They did not agree on our earlier checkpoints, where conditional susceptibility appeared matched under PGD-10 and slightly ViT-favorable under AutoAttack. Since the absolute ordering was identical in both checkpoint sets, a paper reporting only absolute accuracy would have seen nothing change, while a paper reporting the decomposition from a single run would have drawn a mechanistic conclusion that does not replicate. The practical lesson is that mechanism-level claims need more training runs than ranking-level claims, and the two should not be given the same evidential weight.

Sparsity is not spatial locality. Adversarially trained CNN gradients concentrate more mass on fewer components (Hoyer 0.4928 versus 0.4561, paired $p < 10^{-11}$), but four spatial descriptors of the same gradients show no corresponding difference in how those components are arranged in the image. Descriptions of CNN gradients as spatially localized therefore need separate evidence; sparsity measures [61] do not provide

it.

Adversarial perturbation does not measurably change CLS attention entropy. At $n = 1000$ the per-layer entropy change is at most 0.005 nats, while attention maps are displaced by a small and depth-increasing amount. Attention-shift results obtained with patch attacks [31], [32] concern a different threat model and should not be generalized to L_∞ -bounded perturbations without measurement.

A ViT-specific transfer attack does not improve ViT-sourced transfer here. TGR [12] costs 14.9 points of white-box strength and returns none of it at the target in our regime. We report this as a regime-specific negative result, since our targets are adversarially trained and our data are CIFAR-scale, and it is exactly the kind of claim that a single untested assumption would otherwise import into a comparison.

Two behavioral differences reproduce across independent checkpoint sets and are unaffected by the protocol question. Gradient structure differs: CNN input gradients are sparser under all three scale-invariant measures, adversarial training rather than architecture creates that ordering (the clean-trained pair has it reversed), and cross-sample alignment is higher for the ViT in absolute cosine while the signed mean is near zero for both, so that samples share sensitivity axes, not directions, which is the distinction that separates our measurement from a universal-perturbation claim [63]. Depth profiles differ: adversarial damage in the CNN is a late, localized, magnitude-changing event confined to the last residual stage, whereas in the ViT it is a distributed, direction-only rotation. That the ViT’s measured profile depends on token aggregation, since the CLS pathway drifts three times as far as the patch-token mean, is itself a reporting requirement for layer-wise studies.

B. Reporting Requirements

The measurements above imply a short and concrete set of requirements for cross-architecture robustness studies.

Studies should state the conditioning protocol and report which samples enter the denominator. Where a directional claim is made, a protocol that scores both directions on identical samples should be preferred, so that paired inference is available. Unconditioned transfer rates should not be reported as architecture comparisons. When targets differ in clean accuracy, the raw rate is a known mixture of transferability and target weakness, $r_{\text{raw}} - r_{\text{cond}} = e(1 - r_{\text{cond}})$, and because this is an identity rather than a fitted relation the correction can be applied to published raw rates without re-running anything.

Studies should also report the decomposition and not only the score. Absolute robustness factors exactly into clean accuracy and conditional susceptibility, and which factor carries a headline is the interesting part, which is not recoverable from the score alone. Ranking claims should be separated from mechanism claims: our rankings were stable across checkpoint sets while our mechanism attribution was not, so mechanism claims should carry more training runs. For layer-wise analyses the aggregation should be stated, since CLS-only, patch-mean, and all-token summaries of the same blocks give different drift magnitudes and different minima. Finally, attention and locality claims should be checked at a realistic

sample size, because both of the null results we report would have looked positive on a small figure batch.

For practitioners choosing between the two families under matched standard adversarial training on this benchmark, the CNN gives higher absolute robust accuracy and the advantage holds under every view we measured. The WideResNet reference at 62.76% shows that recipe, capacity, and extra data jointly [54] move robustness further than the architectural choice does, which is worth weighing before investing in architecture selection as a robustness strategy.

C. Limitations

Our study is deliberately narrow, and the narrowness bounds the conclusions in specific ways.

The protocol-sensitivity result is measured on three datasets and one model pair. On CIFAR-100 the spread is close to the CIFAR-10 value (13.83 ± 1.30 against 15.01 ± 0.84 points) and the sign is preserved in all twelve measurements (Section IV-C). The effect is therefore not CIFAR-specific, but its size does not follow the clean-accuracy gap across datasets: CIFAR-100 has by far the larger gap and the slightly smaller spread. The mechanism we propose predicts that a gap produces a spread, not how the spread orders between datasets. That account is incomplete, and we state the limit rather than the slogan: it holds for three of the four protocols, while the successful-source protocol adds a second driver tied to the *source*'s robustness that is strong enough to reverse the overall relationship (Section IV-B). What we still cannot state is the functional form of either dependence with confidence: the estimates rest on six seed-and-dataset clusters, and every small-gap pair in our pool involves the external WideResNet, so a small clean-accuracy gap and a different training recipe are not separable here. A study spanning several architecture pairs with varying clean-accuracy gaps would turn these measurements into a calibration curve.

We cannot isolate the cause of the differences between our two checkpoint sets. The corrected protocol changed the validation handling and produced a fresh training run at the same time. A data-quantity explanation is ruled out, since the corrected clean models are better despite less data, but distinguishing selection leakage from ordinary run-to-run variation would require an ablation that trains on identical data with and without a leaked selection split. We report the contrast as evidence that single-run mechanism claims are fragile, not as a measurement of leakage.

Every model in this paper is trained inside a single threat model: L_∞ at $\epsilon = 8/255$. We evaluated the same checkpoints under an L_2 budget and the protocol spread survived, at 10.92 against 15.01 points (Section IV-E), but that test probes the measurement axis rather than L_2 robustness, and its absolute values are not comparable with L_2 -trained references. Whether the effect holds for models *trained* under another norm is untested here.

Three runs per architecture bound seed variance but do not span the recipe space, and ViT adversarial training is known to be recipe-sensitive on small datasets [56], [57]; with diffusion-generated synthetic data ViTs have been reported

to surpass WideResNets on CIFAR-10 [36], [64]. Our ViT results characterize baseline adversarial training, not the best achievable ViT robustness. The ViT-Tiny pipeline upsamples 32×32 inputs to 224×224 inside the model; our CIFAR-native control shows the gradient ordering is not an upsampling artifact, but that control differs in capacity and patch size and therefore calibrates the confound rather than providing a second matched pair. The two models also differ in parameter count (11.2M versus 5.7M), so architectural attribution of the absolute gap remains partial. Finally, we evaluate a single-budget L_∞ threat model, and our TGR result is an adaptation of the original method rather than a reproduction of its published setting.

D. Future Directions

The most direct extension is to turn the protocol-sensitivity measurement into a general one: sweep architecture pairs with controlled clean-accuracy gaps and measure how the spread across protocols scales with that gap. If the relationship is as regular as our three-target correlation suggests, it would yield a correction that lets published raw-rate asymmetries be reinterpreted rather than discarded.

Two of the extensions we previously listed as future work have since been run, and we record both what they settled and what they did not. The leakage ablation was carried out on six full-budget trajectories, and it did not detect an effect of leakage on checkpoint selection; it answers a weaker question than the one originally posed, because the treatment was applied to a single validation split, so split identity and leakage cannot be separated within that design. Its informative by-product is the selection-protocol dispersion reported in Section IV-I. The second dataset has also been run (Section IV-C). What remains open is the capacity-matched pair (for example ViT-Small against ResNet-50), which would test whether the surviving behavioral differences, the sparsity ordering and the depth profiles, are properties of the families or of these particular small models.

VI. CONCLUSION

We asked how much of the disagreement in CNN and Vision Transformer robustness comparisons comes from the measurement rather than from the models, and answered it by holding the models, the data, the threat model, and the attack budget fixed while varying only how the comparison is scored.

The measurement dominates. On identical adversarially trained ResNet-18 and ViT-Tiny pairs, the cross-architecture transfer asymmetry ranges from +4.4 to +19.4 points across four established conditioning protocols, a 4.4-fold spread, whereas retraining both architectures from scratch moves that same quantity by under 1.5 points, and the protocol decides whether an equivalence test is accepted or rejected. Unconditioned rates are not a valid basis for such a comparison at all: in a 3×3 matrix that adds a WideResNet-28-10 reference, their deviation from conditioned rates is algebraically equal to the target's clean error scaled by one minus the conditioned rate, an identity that holds to within 0.41 points across 36 measured directions and that applies to any published raw

rate whose targets satisfy the measured antecedent. The same design run on CIFAR-100 leaves the spread at 13.83 ± 1.30 points, close to the CIFAR-10 value, and an ablation that fixes the trajectory and varies only the checkpoint-selection rule moves reported robust accuracy by 1.58 to 2.85 points, so measurement choices that are usually left implicit are of the same order as the effects being reported. The direction is stable across protocols, seeds, and attack families on both CIFAR datasets, but not across datasets: on SVHN, where the same recipe leaves the two architectures only 1.85 points apart in clean accuracy, two of the four protocols reverse the sign and the primary protocol accepts equivalence. What survives is narrower and more useful than a ranking: the direction holds while the models are unmatched, and our three-target analysis associates it with the target’s own vulnerability ($r = 0.986$) rather than with any special potency of CNN-crafted perturbations.

A conditional decomposition separates absolute robustness into a clean-accuracy factor and a conditional-susceptibility factor. Both favor the CNN in our corrected runs, but the attribution between them differed on an earlier checkpoint set whose absolute ordering was identical, which shows that mechanism-level claims are more fragile than ranking-level claims and need more training runs to support.

Three claims common in this literature did not survive direct measurement: adversarially trained CNN gradients are sparser but not spatially more localized, CLS attention entropy does not change under L_∞ attack at $n = 1000$, and a transformer-specific transfer attack yields no transfer gain in this regime. Two behavioral differences did survive and reproduce across independent checkpoint sets: the gradient sparsity ordering, which adversarial training creates rather than architecture, and the depth profile of adversarial damage, which is a late localized magnitude-changing event in the CNN and a distributed direction-only rotation in the ViT.

From these measurements we distilled a short set of reporting requirements, of which the first two matter most: state the conditioning protocol and prefer paired ones, and do not present unconditioned transfer rates as architecture comparisons. Our study covers three datasets and a single model pair, so the size of the protocol effect for other architecture pairs remains to be calibrated; the mechanism behind it, an unequal clean-accuracy gap between targets, is not specific to our setting. The pipeline, the per-sample logs and the analysis scripts will be released on acceptance so that the protocol comparison can be reproduced and extended.

DATA AND CODE AVAILABILITY

The experiments in this study use the CIFAR-10 and CIFAR-100 datasets [11] and the SVHN dataset [62], all of which are publicly available. Pre-trained WideResNet-28-10 weights were obtained from the RobustBench benchmark [23], [54]. Upon acceptance we will publish the analysis pipeline, the scripts that generate every table and figure in this paper, the per-sample evaluation records, the fixed validation-split indices and the seed lists, which together reproduce every number reported here; the four conditioning protocols are all computed

from the per-sample records, so the protocol comparison can be repeated without retraining. Trained checkpoints total about 36 GB and are available from the authors on request.

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REFERENCES

- [1] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, “Intriguing properties of neural networks,” in *International Conference on Learning Representations*, 2014.
- [2] I. J. Goodfellow, J. Shlens, and C. Szegedy, “Explaining and harnessing adversarial examples,” in *International Conference on Learning Representations*, 2015.
- [3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations*, 2021.
- [4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
- [6] Y. Bai, J. Mei, A. L. Yuille, and C. Xie, “Are transformers more robust than cnns?” in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 26 831–26 843.
- [7] P. Benz, S. Ham, C. Zhang, A. Karjauv, and I. S. Kweon, “Adversarial robustness comparison of vision transformer and MLP-Mixer to CNNs,” in *Proceedings of the 32nd British Machine Vision Conference (BMVC)*, 2021.
- [8] K. Mahmood, R. Mahmood, and M. Van Dijk, “On the robustness of vision transformers to adversarial examples,” in *IEEE/CVF International Conference on Computer Vision*, 2021, pp. 7838–7847.
- [9] D. Ravikumar, S. Kodge, I. Garg, and K. Roy, “TREND: Transferability-based robust ENsemble design,” *IEEE Transactions on Artificial Intelligence*, vol. 4, no. 3, pp. 534–548, 2023.
- [10] M. Naseer, K. Ranasinghe, S. Khan, F. S. Khan, and F. Porikli, “On improving adversarial transferability of vision transformers,” in *International Conference on Learning Representations*, 2022.
- [11] A. Krizhevsky, G. Hinton *et al.*, “Learning multiple layers of features from tiny images,” University of Toronto, Tech. Rep., 2009.
- [12] J. Zhang, Y. Huang, W. Wu, and M. R. Lyu, “Transferable adversarial attacks on vision transformers with token gradient regularization,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 16 415–16 424.
- [13] c. Şahin and O. Yıldırım, “A behavioral comparison of adversarially trained ResNet-18 and ViT-Tiny on CIFAR-10: Transfer metrics, gradient structure, and block-wise feature drift,” in *Proc. ATEEC 2026*, 2026, preliminary version of the present study; submitted.
- [14] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, “Towards deep learning models resistant to adversarial attacks,” in *International Conference on Learning Representations*, 2018.
- [15] N. Carlini and D. Wagner, “Towards evaluating the robustness of neural networks,” in *IEEE Symposium on Security and Privacy (SP)*, 2017, pp. 39–57.
- [16] F. Croce and M. Hein, “Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks,” in *International Conference on Machine Learning*, 2020, pp. 2206–2216.
- [17] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. El Ghaoui, and M. Jordan, “Theoretically principled trade-off between robustness and accuracy,” in *International Conference on Machine Learning*, 2019, pp. 7472–7482.
- [18] Y. Wang, D. Zou, J. Yi, J. Bailey, X. Ma, and Q. Gu, “Improving adversarial robustness requires revisiting misclassified examples,” in *International Conference on Learning Representations*, 2020.
- [19] C. Guo, M. Rana, M. Cisse, and L. Van Der Maaten, “Countering adversarial images using input transformations,” in *International Conference on Learning Representations*, 2018.
- [20] J. Cohen, E. Rosenfeld, and Z. Kolter, “Certified adversarial robustness via randomized smoothing,” in *International Conference on Machine Learning*, 2019, pp. 1310–1320.

- [21] F. Tramèr, A. Kurakin, N. Papernot, I. Goodfellow, D. Boneh, and P. McDaniel, “Ensemble adversarial training: Attacks and defenses,” in *International Conference on Learning Representations*, 2018.
- [22] S. Lyu, S. Shaikh, F. Shpilevskiy, E. Shelhamer, and M. Lécuyer, “Adaptive randomized smoothing: Certified adversarial robustness for multi-step defences,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024.
- [23] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, and M. Hein, “RobustBench: a standardized adversarial robustness benchmark,” in *NeurIPS Datasets and Benchmarks Track*, 2021.
- [24] D. Wu, Y. Wang, S.-T. Xia, J. Bailey, and X. Ma, “Skip connections matter: On the transferability of adversarial examples generated with ResNets,” in *International Conference on Learning Representations (ICLR)*, 2020.
- [25] C. Xie and A. Yuille, “Intriguing properties of adversarial training at scale,” in *International Conference on Learning Representations (ICLR)*, 2020.
- [26] C. Xie, M. Tan, B. Gong, J. Wang, A. L. Yuille, and Q. V. Le, “Adversarial examples improve image recognition,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 819–828.
- [27] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, and A. Madry, “Robustness may be at odds with accuracy,” in *International Conference on Learning Representations (ICLR)*, 2019.
- [28] P. Chalasani, J. Chen, A. R. Chowdhury, X. Wu, and S. Jha, “Concise explanations of neural networks using adversarial training,” in *Proceedings of the 37th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 119. PMLR, 2020, pp. 1383–1391.
- [29] M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, and A. Dosovitskiy, “Do vision transformers see like convolutional neural networks?” in *Advances in Neural Information Processing Systems*, vol. 34. Curran Associates, Inc., 2021, pp. 12 116–12 128.
- [30] S. Bhojanapalli, A. Chakrabarti, D. Glasner, D. Li, T. Unterthiner, and A. Veit, “Understanding robustness of transformers for image classification,” in *IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10 231–10 241.
- [31] Y. Fu, S. Zhang, S. Wu, C. Wan, and Y. Lin, “Patch-Fool: Are vision transformers always robust against adversarial perturbations?” in *International Conference on Learning Representations (ICLR)*, 2022.
- [32] J. Gu, V. Tresp, and Y. Qin, “Are vision transformers robust to patch perturbations?” in *Computer Vision – ECCV 2022*, ser. Lecture Notes in Computer Science, vol. 13672. Springer, 2022, pp. 404–421.
- [33] R. Shao, Z. Shi, J. Yi, P.-Y. Chen, and C.-J. Hsieh, “On the adversarial robustness of vision transformers,” *Transactions on Machine Learning Research*, 2022.
- [34] S. Paul and P.-Y. Chen, “Vision transformers are robust learners,” in *AAAI Conference on Artificial Intelligence*, 2022, pp. 2071–2081.
- [35] S. Jain and T. Dutta, “Towards understanding and improving adversarial robustness of vision transformers,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 24 736–24 745.
- [36] J. Wu, Z. Song, X. Zhang, S. Xie, L. Lin, and K. Wang, “Vision transformers beat WideResNets on small scale datasets adversarial robustness,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 1, 2025, pp. 886–894.
- [37] Z. Zhu, X. Wang, Z. Jin, J. Zhang, and H. Chen, “Enhancing transferable adversarial attacks on vision transformers through gradient normalization scaling and high-frequency adaptation,” in *International Conference on Learning Representations*, 2024.
- [38] N. Papernot, P. McDaniel, and I. Goodfellow, “Transferability in machine learning: from phenomena to black-box attacks using adversarial samples,” *arXiv preprint arXiv:1605.07277*, 2016.
- [39] A. Demontis, M. Melis, M. Pintor, M. Jagielski, B. Biggio, A. Oprea, C. Nita-Rotaru, and F. Roli, “Why do adversarial attacks transfer? explaining transferability of evasion and poisoning attacks,” in *28th USENIX Security Symposium*, 2019, pp. 321–338.
- [40] Y. Liu, X. Chen, C. Liu, and D. Song, “Delving into transferable adversarial examples and black-box attacks,” in *Proceedings of the 5th International Conference on Learning Representations (ICLR)*, 2017.
- [41] Y. Dong, F. Liao, T. Pang, H. Su, J. Zhu, X. Hu, and J. Li, “Boosting adversarial attacks with momentum,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 9185–9193.
- [42] Z. Zhao, H. Zhang, R. Li, R. Sicre, L. Amsaleg, M. Backes, Q. Li, Q. Wang, and C. Shen, “Revisiting transferable adversarial images: Systemization, evaluation, and new insights,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 48, no. 1, pp. 765–780, 2026.
- [43] Q. Li, Y. Guo, W. Zuo, and H. Chen, “Towards evaluating transfer-based attacks systematically, practically, and fairly,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023.
- [44] W. Yu, J. Gu, Z. Li, and P. Torr, “Reliable evaluation of adversarial transferability,” in *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2025, pp. 797–810.
- [45] Z. Wei, J. Chen, M. Goldblum, Z. Wu, T. Goldstein, and Y.-G. Jiang, “Towards transferable adversarial attacks on vision transformers,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 3, 2022, pp. 2668–2676.
- [46] W. Ma, Y. Li, X. Jia, and W. Xu, “Transferable adversarial attack for both vision transformers and convolutional networks via momentum integrated gradients,” in *IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4630–4639.
- [47] Z. Chen, C. Xu, H. Lv, S. Liu, and Y. Ji, “Understanding and improving adversarial transferability of vision transformers and convolutional neural networks,” *Information Sciences*, vol. 648, p. 119474, 2023.
- [48] F. Waseda, S. Nishikawa, T.-N. Le, H. H. Nguyen, and I. Echizen, “Closer look at the transferability of adversarial examples: How they fool different models differently,” in *IEEE/CVF Winter Conference on Applications of Computer Vision*, 2023, pp. 1360–1368.
- [49] F. Pinto, P. H. S. Torr, and P. K. Dokania, “An impartial take to the CNN vs Transformer robustness contest,” in *Computer Vision – ECCV 2022*, ser. Lecture Notes in Computer Science, vol. 13673. Cham: Springer, 2022, pp. 466–480.
- [50] K. Ali, M. S. Bhatti, A. Saeed, A. Athar, M. A. Al Ghamdi, S. H. Almotiri, and S. Akram, “Adversarial robustness of vision transformers versus convolutional neural networks,” *IEEE Access*, vol. 12, pp. 105 281–105 293, 2024.
- [51] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Madry, and A. Kurakin, “On evaluating adversarial robustness,” *arXiv preprint arXiv:1902.06705*, 2019.
- [52] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, “On adaptive attacks to adversarial example defenses,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [53] S. Zagoruyko and N. Komodakis, “Wide residual networks,” in *Proceedings of the British Machine Vision Conference (BMVC)*. BMVA Press, 2016, pp. 87.1–87.12.
- [54] S. Goyal, C. Qin, J. Uesato, T. Mann, and P. Kohli, “Uncovering the limits of adversarial training against norm-bounded adversarial examples,” *arXiv preprint arXiv:2010.03593*, 2020.
- [55] R. Wightman, “Pytorch image models,” <https://github.com/rwightman/pytorch-image-models>, 2019.
- [56] E. Debenedetti, V. Sehwag, and P. Mittal, “A light recipe to train robust vision transformers,” in *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, 2023, pp. 225–253.
- [57] Y. Mo, D. Wu, Y. Wang, Y. Guo, and Y. Wang, “When adversarial training meets vision transformers: Recipes from training to architecture,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 18 599–18 611.
- [58] L. Rice, E. Wong, and J. Z. Kolter, “Overfitting in adversarially robust deep learning,” in *International Conference on Machine Learning*, 2020, pp. 8093–8104.
- [59] I. Loshchilov and F. Hutter, “SGDR: Stochastic gradient descent with warm restarts,” in *International Conference on Learning Representations (ICLR)*, 2017.
- [60] —, “Decoupled weight decay regularization,” in *International Conference on Learning Representations (ICLR)*, 2019.
- [61] N. Hurley and S. Rickard, “Comparing measures of sparsity,” *IEEE Transactions on Information Theory*, vol. 55, no. 10, pp. 4723–4741, Oct. 2009.
- [62] Y. Netzer, T. Wang, A. Coates, A. Bissacco, B. Wu, and A. Y. Ng, “Reading digits in natural images with unsupervised feature learning,” in *NIPS Workshop on Deep Learning and Unsupervised Feature Learning*, 2011.
- [63] S.-M. Moosavi-Dezfooli, A. Fawzi, O. Fawzi, and P. Frossard, “Universal adversarial perturbations,” in *IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 1765–1773.
- [64] Z. Wang, T. Pang, C. Du, M. Lin, W. Liu, and S. Yan, “Better diffusion models further improve adversarial training,” in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 36 246–36 263.

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